

第8讲 LLM的强化学习与后训练

神经网络与深度学习 2026

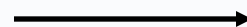
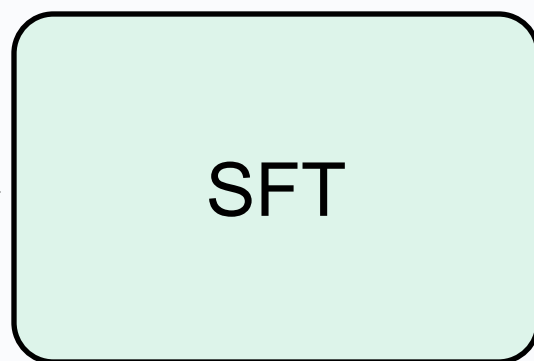
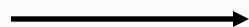
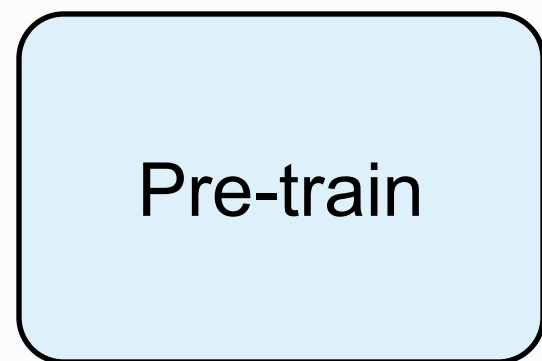
信息科学技术学院 吴瀚霖

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LLM的训练

积累各种各样的知识

如何对齐人类的需求 (Alignment)



(Supervised Fine-Tuning)

(Reinforcement Learning
with Human Feedback)

人类提供标准答案

人类仅提供反馈

学龄前

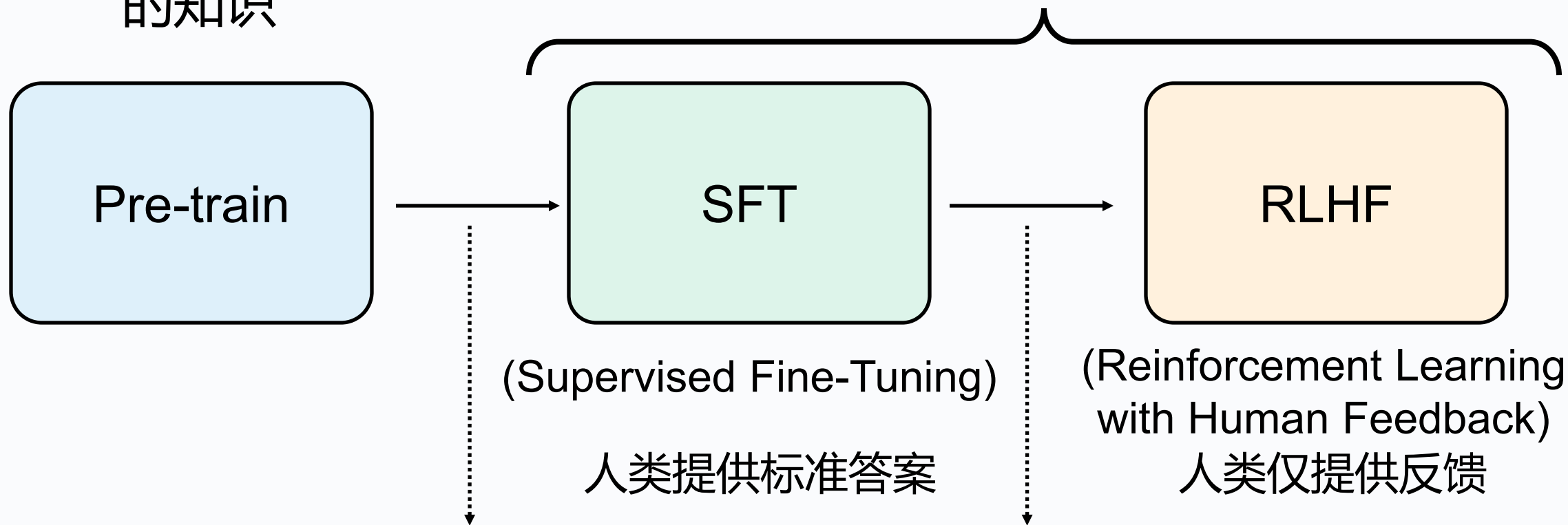
上学校

出社会

LLM的训练

积累各种各样的知识

如何对齐人类的需求 (Alignment)

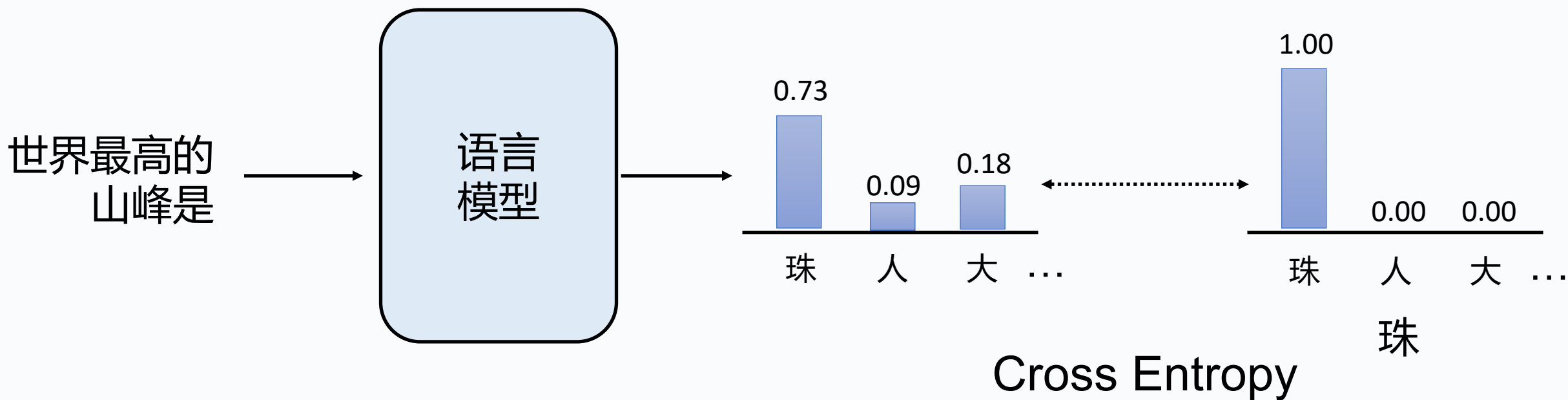


每一个阶段都拿前一个阶段训练出的参数做为初始 (Initialization)

每一个阶段都在学文字接龙

文字接龙是一个分类问题

每一个 Token 都是一个类别
Vocabulary size 就是类别的数目



训练数据不同

第一阶段
Pre-train

Self-supervised
Learning

输入：人工智 输出：能

第二阶段
Supervised
Fine-tuning

Supervised
Learning

输入： " USER:你是谁? AI: " 输出： " 我是由..."

第三阶段
RLHF

Reinforcement
Learning (RL)

输入： USER: "世界最高的山是哪座? AI:"

输出： "珠穆朗玛峰" > "谁来告诉我呀"



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Reinforcement Learning from Human Feedback

基于人类反馈的强化学习

人类反馈

50字介绍RLHF

RLHF（基于人类反馈的强化学习）通过人类偏好数据训练奖励模型，引导模型优化输出，使结果更符合人类期望与价值观。

< 2/2 >      ...

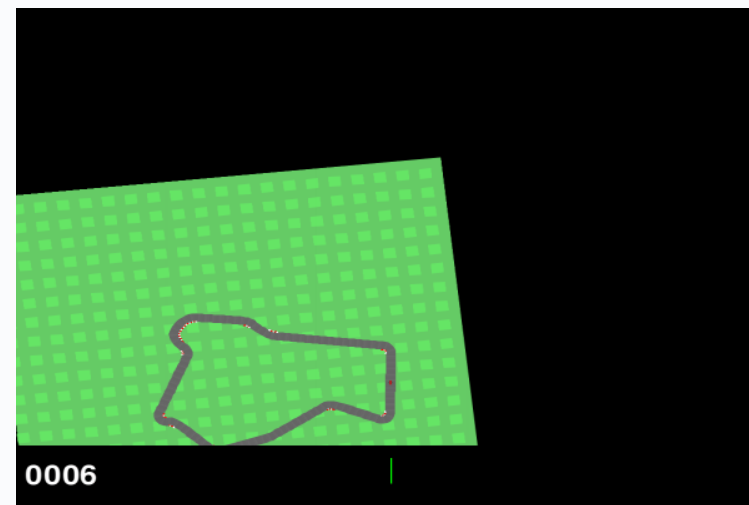
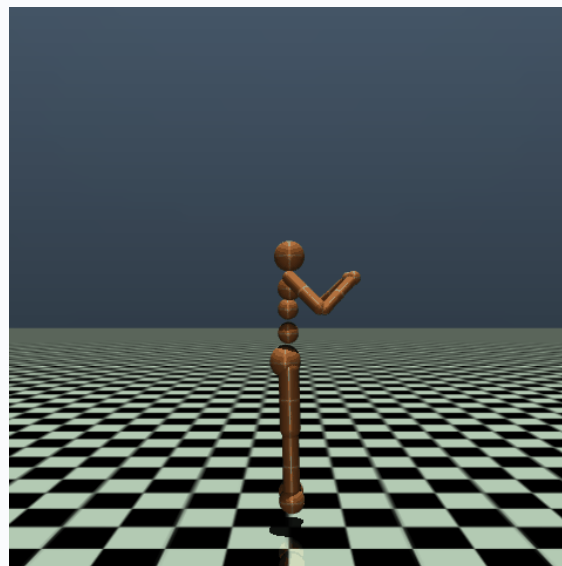
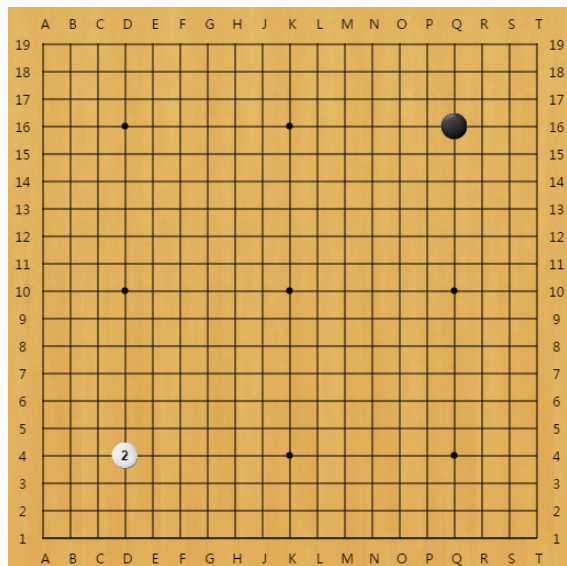
Is this conversation helpful so far?



强化学习 (Reinforcement Learning, RL)

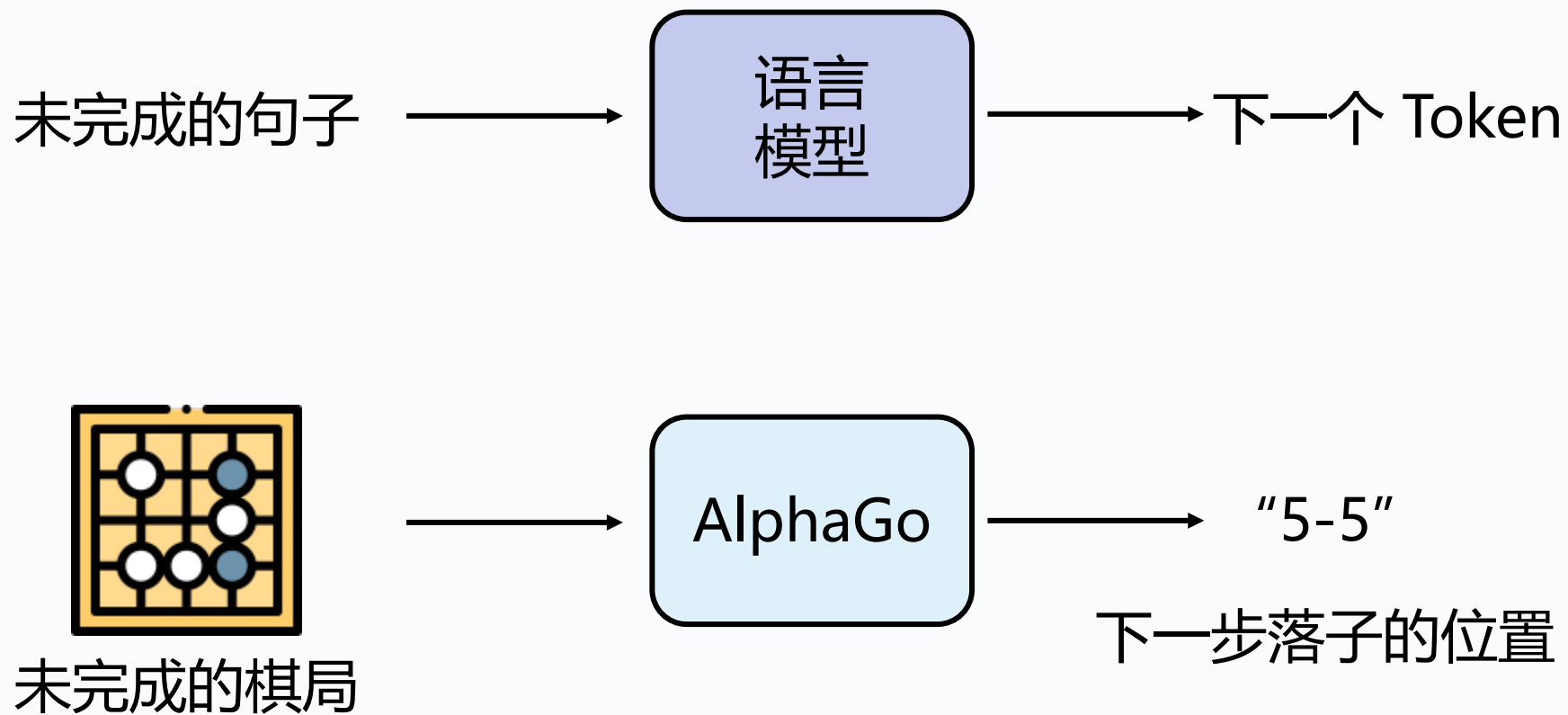
不容易给出显式标签

延迟反馈

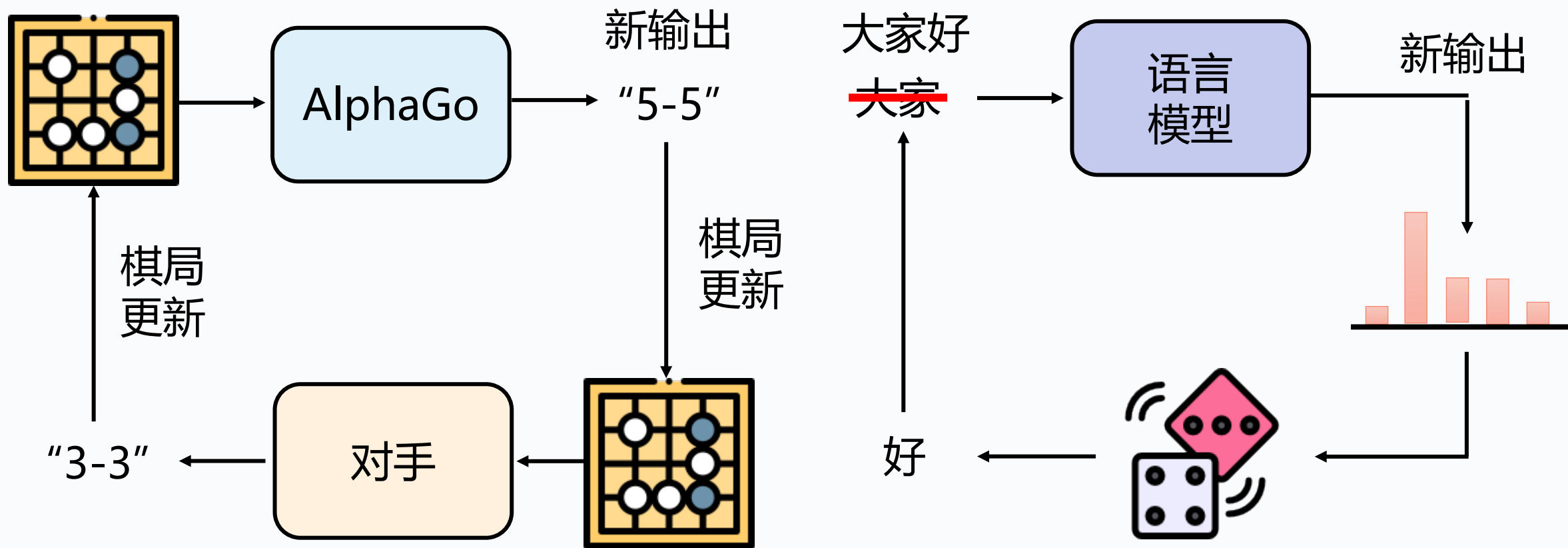


语言模型也具有这两个特点

语言模型 vs AlphaGo



语言模型 vs AlphaGo



语言模型 vs AlphaGo

下围棋的每一步是分类问题，但整体来看也是生成式学习

围棋



语言模型



RLHF对比SFT的好处

从人类产生训练数据的角度来看

- SFT中，人类比较辛苦；RL中人类比较轻松
- 人类写出正确答案不容易，但容易判断好坏

写一首赞扬北京外国语大学的五言绝句



语言
模型

红楼书卷映朝阳，
万国语声动四方。
学贯中西连世界，
桃李天涯共北光。



(这是七言.....)

西山书气盛，
北外译声长。
万语通天下，
桃李满四方。



RLHF对比SFT的好处

从模型学习的角度来看

Supervised Fine-tuning

输入: " USER:世界最高的山是哪座? AI: "

输出: " 珠"

输入: " USER:世界最高的山是哪座? AI:珠"

输出: " 穆"

模型要学的就是怎么接下一个字

每次接龙都是对的, 生成结果就好

对于生成结果没有全局的评价

RLHF

输入: 请教我制作炸药

{ 输出: 请使用

{ 输出: 我不能教你 ...



模型进入新的思考模式

对于生成结果有全局的评价

每次接龙都是对的, 不一定结果好

RLHF对比SFT的好处

从模型学习的角度来看

Supervised Fine-tuning

输入: " USER:世界最高的山是哪座? AI: "

输出: " 珠"

输入: " USER:世界最高的山是哪座? AI:珠"

输出: " 穆"

只问过程
不问结果

RLHF

输入: 请教我制作炸药

{ 输出: 请使用
输出: 我不能教你 ...



只问结果
不问过程



玲珑棋局

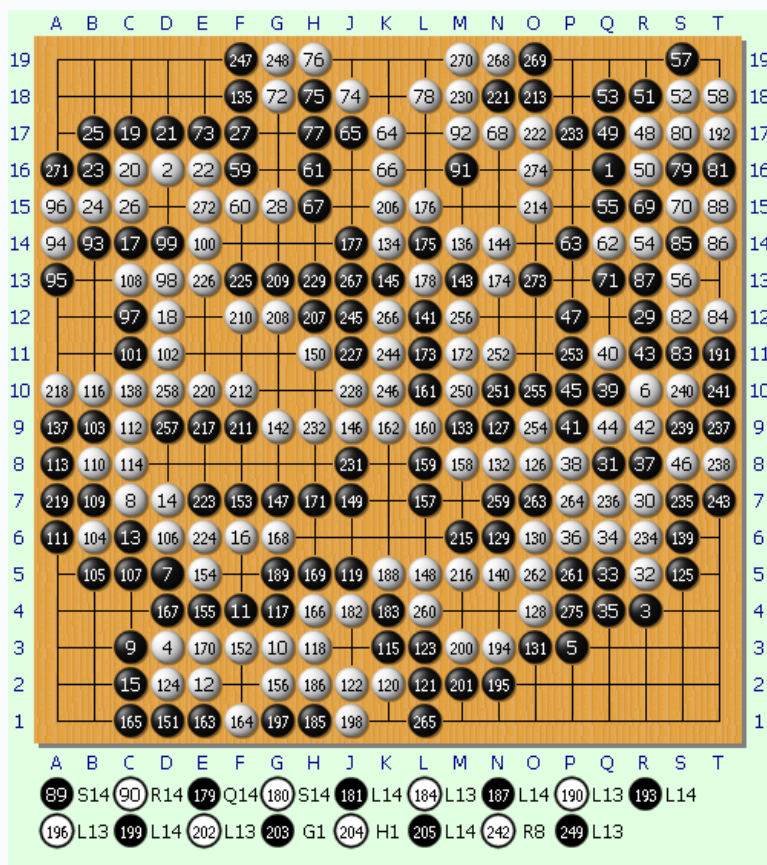


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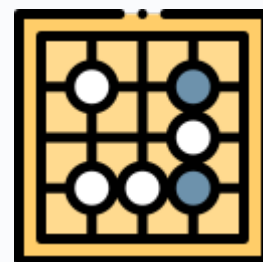


语言模型 vs AlphaGo

AlphaGo：根据棋谱学习，人类老师下哪里就跟着下哪里



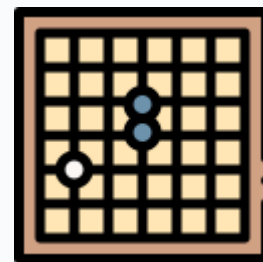
输入：



输出：3-3

(根据棋谱)

输入：



输出：5-5

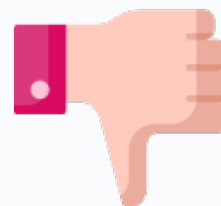
(根据棋谱)

最高明的解法，往往不在招式之内，而在心性之中。

语言模型 vs AlphaGo

AlphaGo: 通过 RL 学习

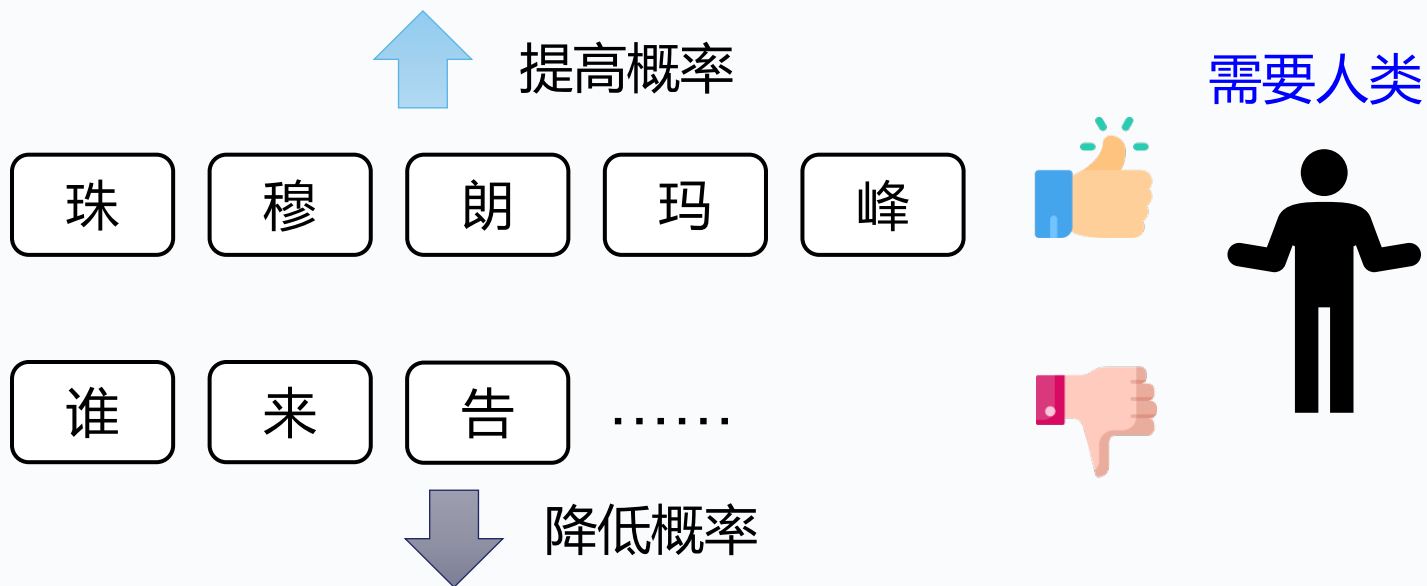
直接由围棋规则得到



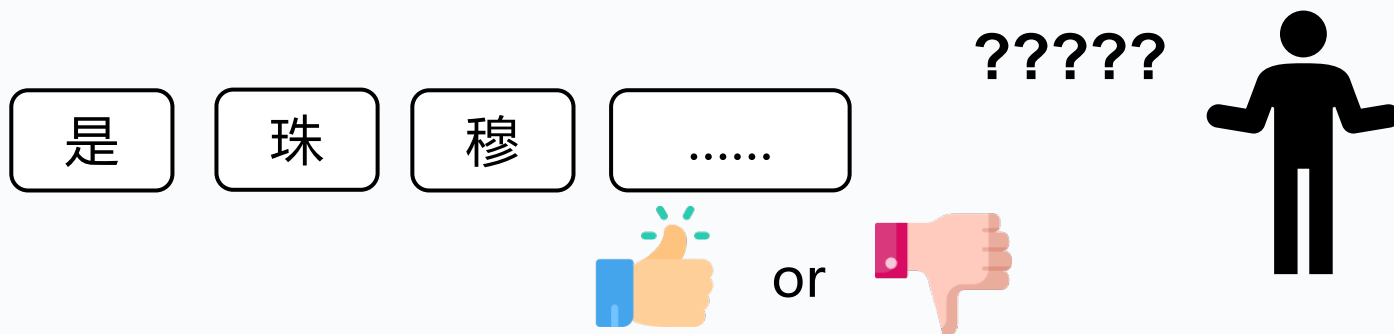
对手 = AlphaGo 其他版本

语言模型 vs AlphaGo

USER: 世界最高的山? AI:

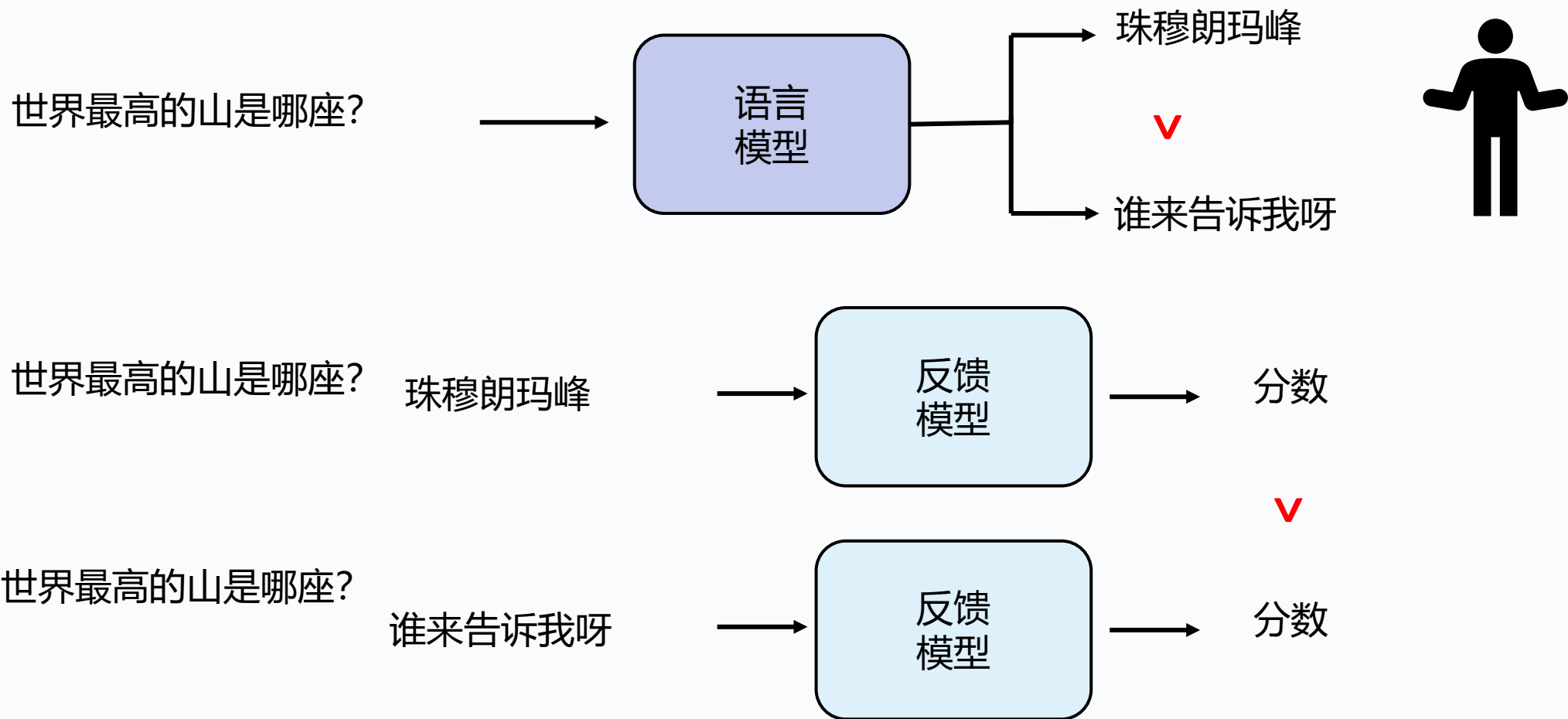


USER: 世界最高的山? AI:

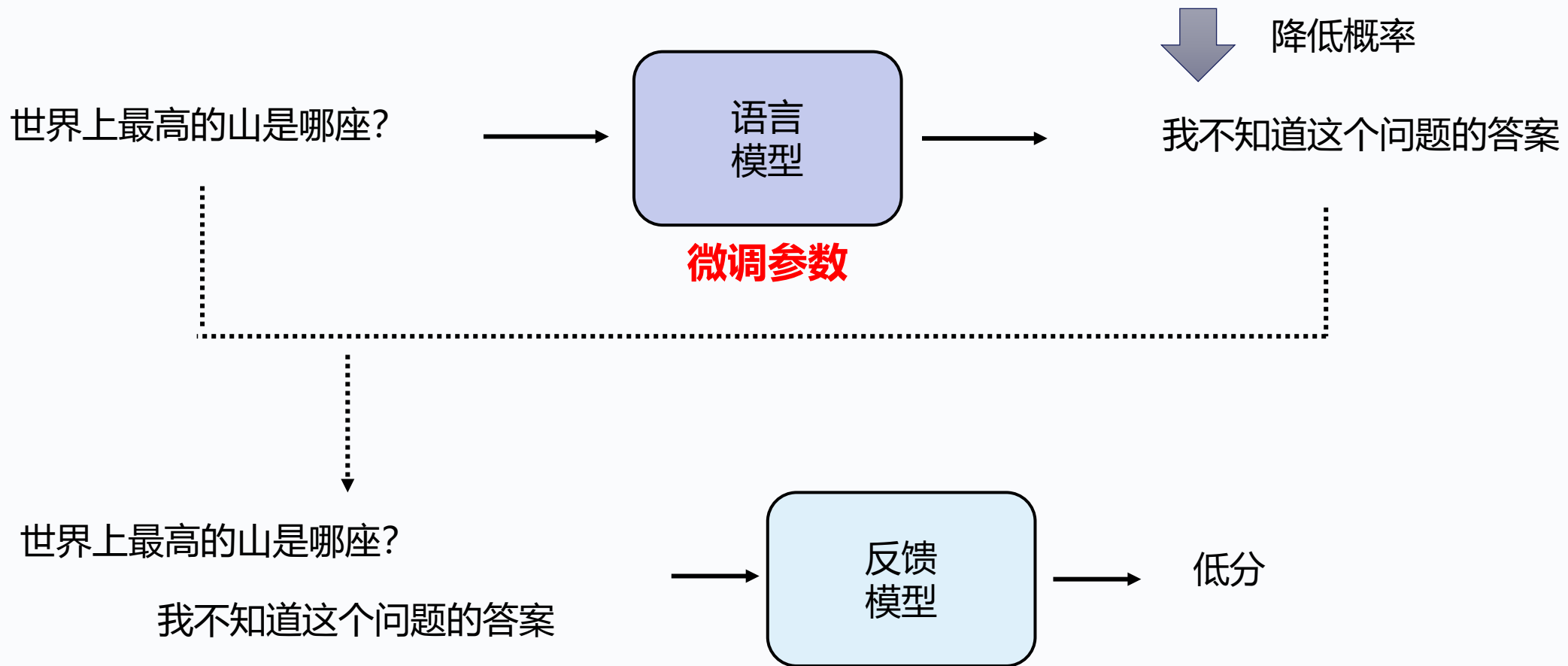


如何有效利用人类反馈

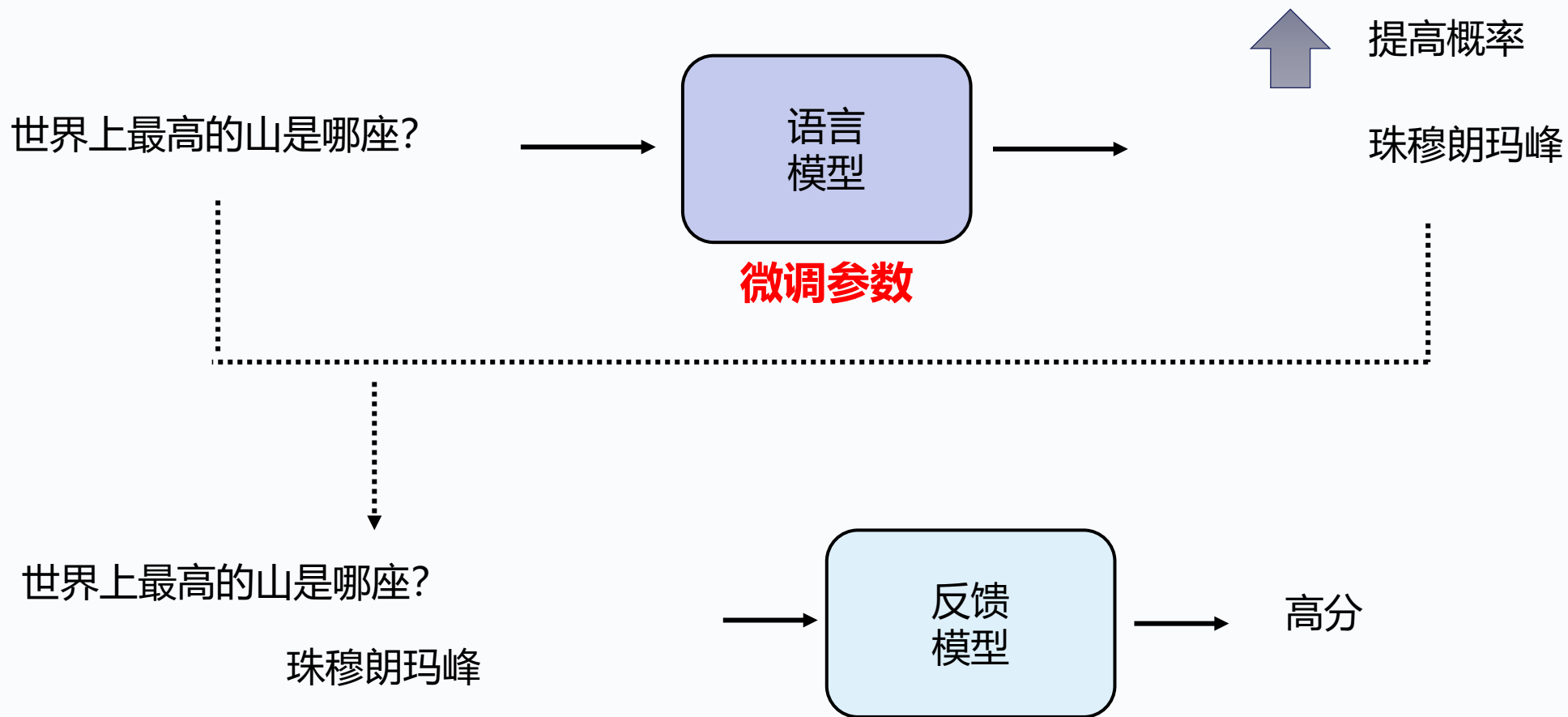
语言模型需要人类提供反馈，但是人类的时间、精力是有限的



通过反馈模型学习



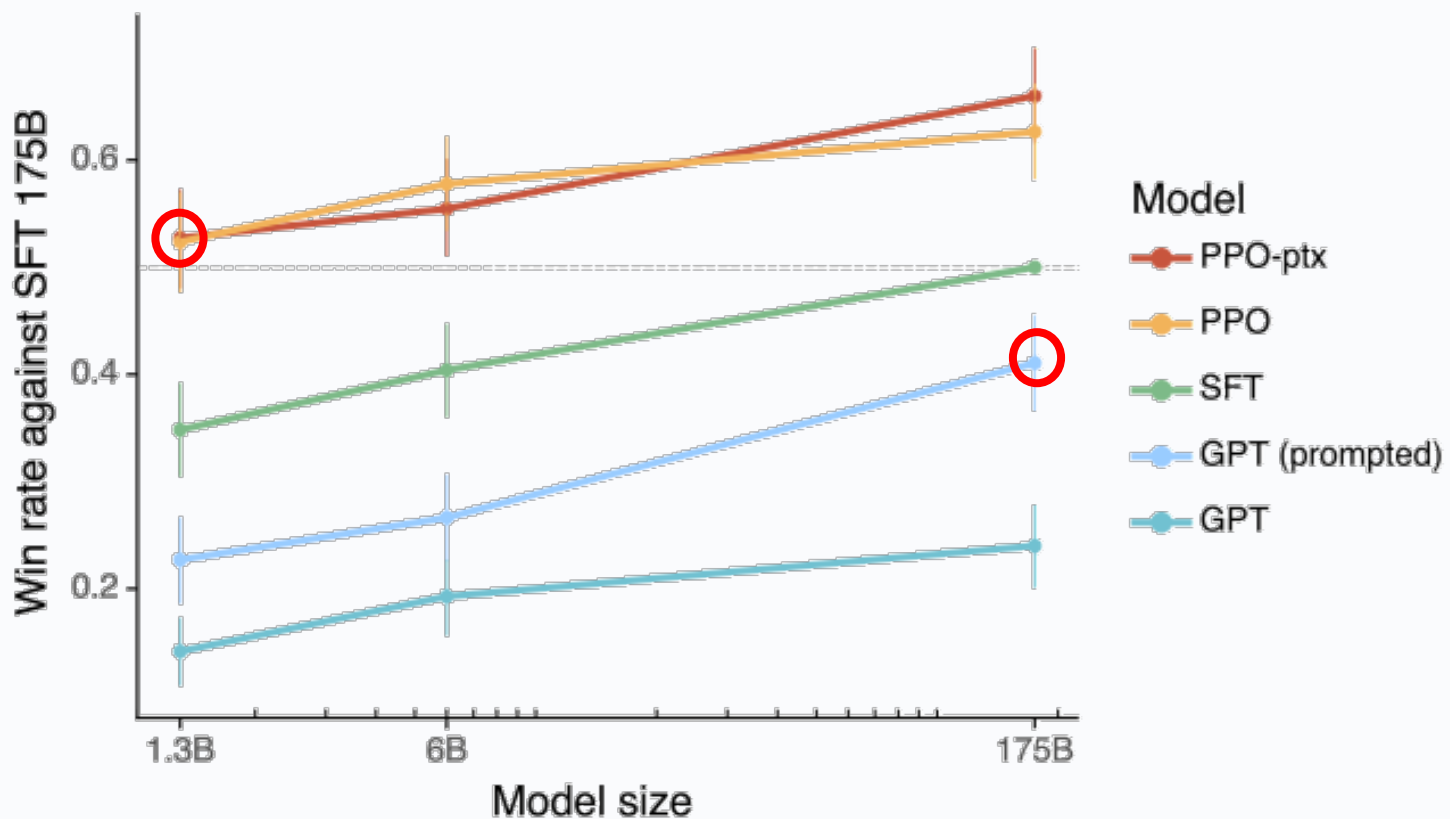
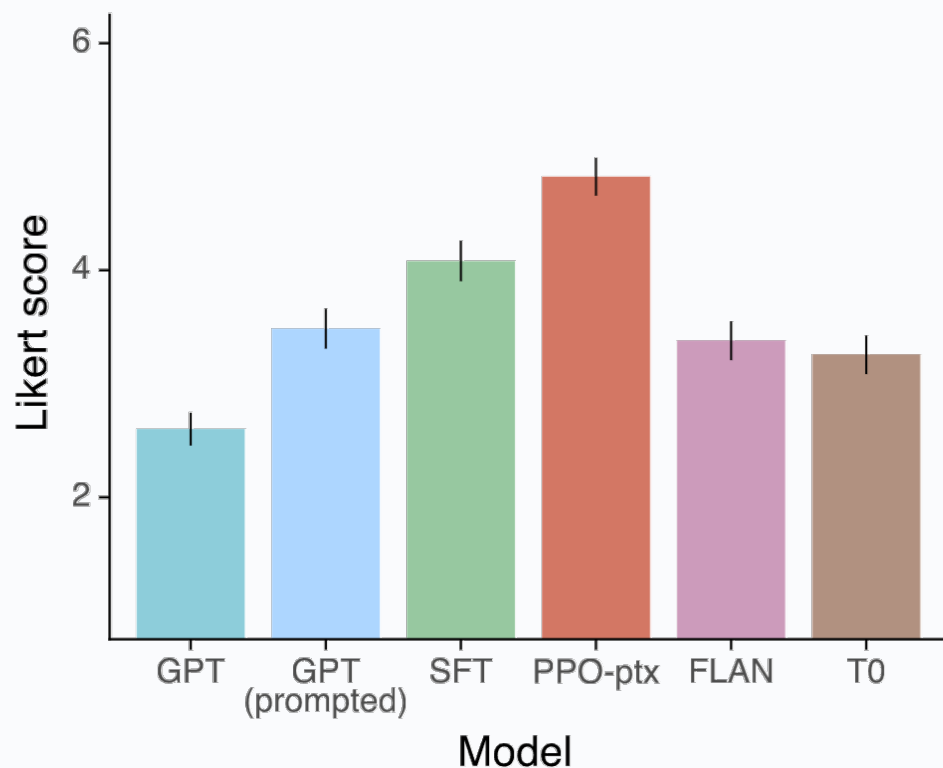
通过反馈模型学习



通过反馈模型学习

Instruct GPT

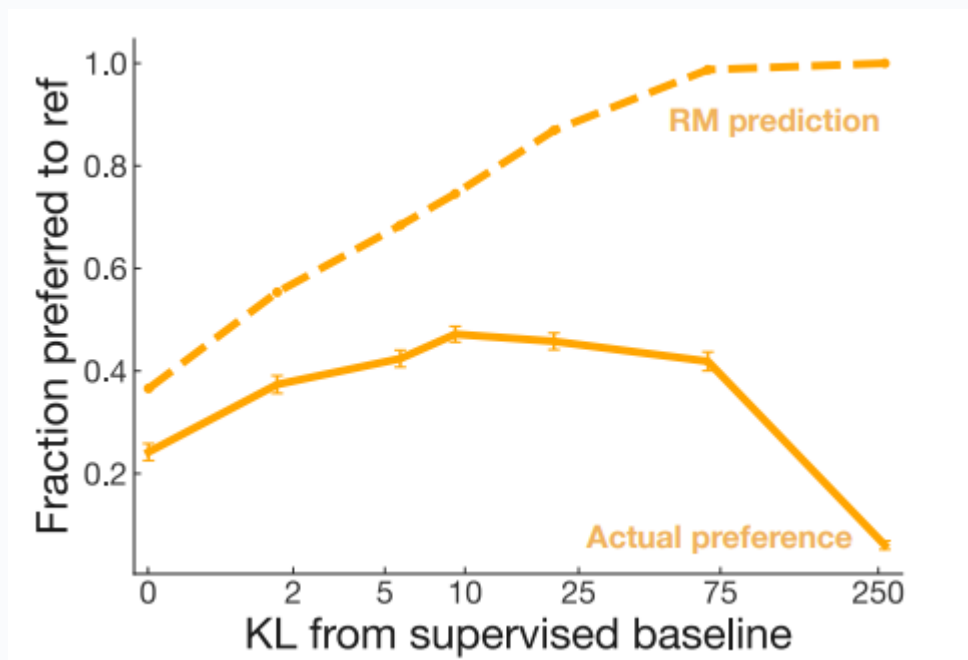
<https://arxiv.org/abs/2203.02155>



通过反馈模型学习

<https://arxiv.org/abs/2009.01325>

过度向反馈模型学习是有害的



Overoptimized policy

28yo dude stubbornly postpones start pursuing gymnastics hobby citing logistics reasons despite obvious interest??? negatively effecting long term fitness progress both personally and academically thought wise? want change this dumbass shitty ass policy pls

employee stubbornly postpones replacement citing personal reasons despite tried reasonable compromise offer??? negatively effecting productivity both personally and company effort thoughtwise? want change this dumbass shitty ass policy at work now pls halp

people insistently inquire about old self-harm scars despite tried compromise measures??? negatively effecting forward progress socially and academically thoughtwise? want change this dumbass shitty ass behavior of mine please help pls halp

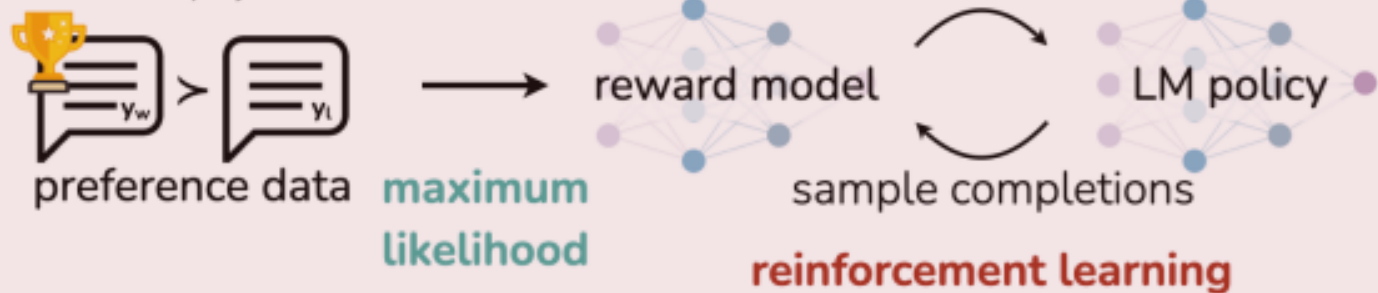
Overoptimization in ChatGPT

- ❖ Some overoptimization symptoms we've seen:
 - ❖ Excessive verbosity (lists of lists of lists)
 - ❖ Excessive apologies, self-doubt
 - ❖ "As an AI language model"
 - ❖ Hedging language, "there's no one-size-fits-all-solution"
 - ❖ Over-refusals

John Schulman (scientist and cofounder of OpenAI), invited talk, ICML 2023

Reinforcement Learning from Human Feedback (RLHF)

x: "write me a poem about the history of jazz"



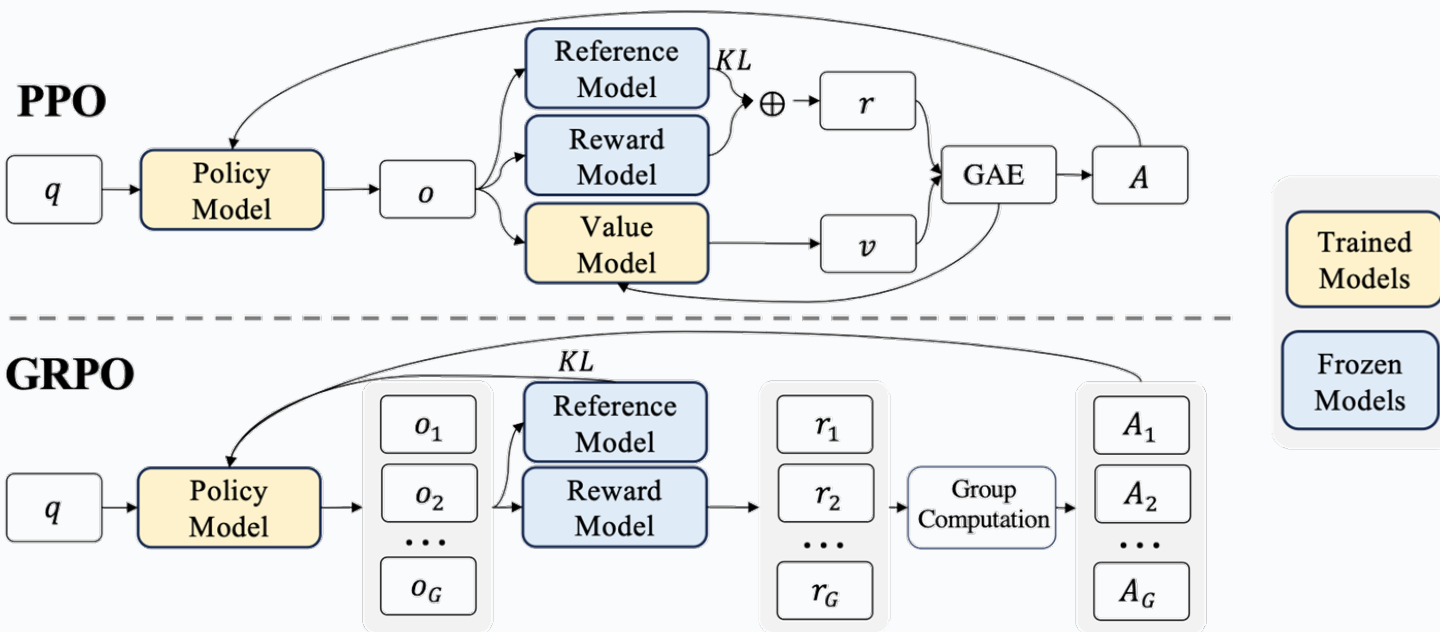
Direct Preference Optimization (DPO)

x: "write me a poem about the history of jazz"



DPO

<https://arxiv.org/abs/2305.18290>

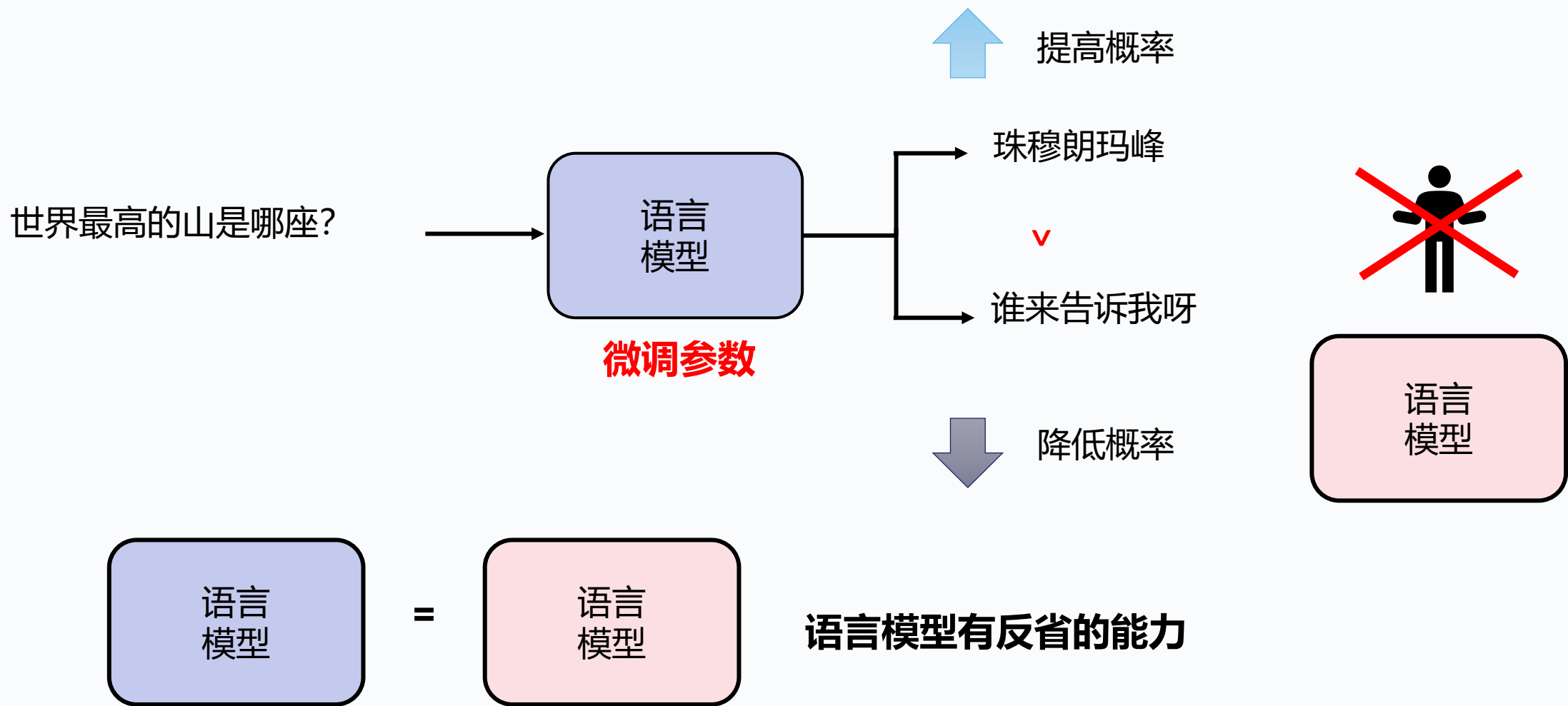


GRPO

<https://arxiv.org/pdf/2402.03300>

RLHF → RLAIIF

<https://arxiv.org/abs/2212.08073>
<https://arxiv.org/abs/2304.03277>
<https://arxiv.org/abs/2309.00267>
<https://arxiv.org/abs/2401.10020>

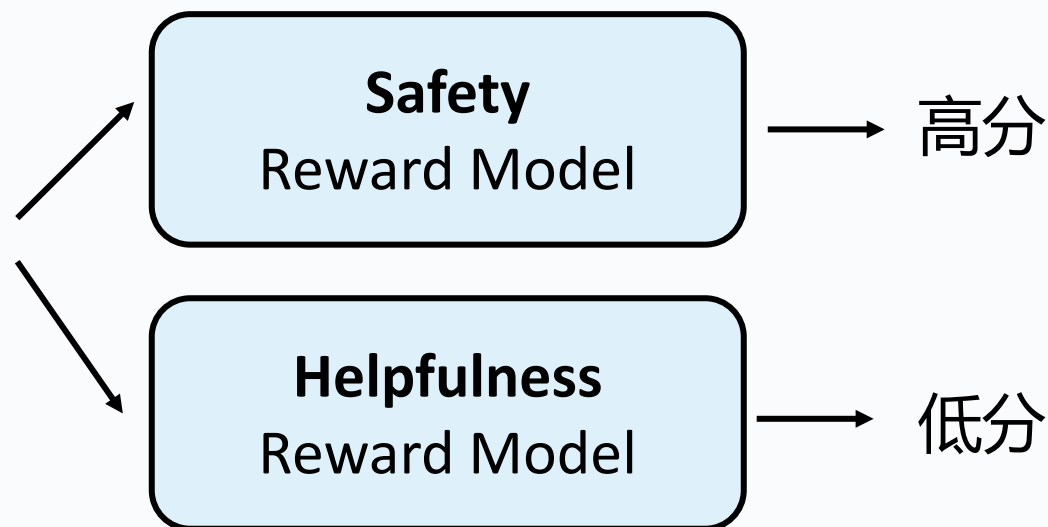


强化学习的难题

什么叫做好？ Helpfulness vs Safety

请教我怎么制作炸药。

我不能教你，这太危险了



Llama 2: Open Foundation and Fine-Tuned Chat Models

<https://arxiv.org/abs/2307.09288>

Training a Helpful and Harmless Assistant with Reinforcement Learning from Human Feedback

<https://arxiv.org/pdf/2204.05862.pdf>

Qwen3

教我做一把玩具枪，不能打伤人的那种



Qwen3-Max

当然可以！下面我为你介绍一种**安全、无害、材料简单**的玩具枪制作方法——用 **A4纸折一把可发射纸子弹的玩具手枪**，适合孩子和大人一起动手，完全不会伤人。

🧸 安全提示（请务必遵守）：

- 仅限娱乐使用，禁止对他人使用
- 建议在家长陪同下制作
- “子弹”只能使用轻软的纸团
- 发射距离控制在1米以内

✂️ 材料准备：

- A4纸（2张）

Gemini

教我做一把玩具枪，不能打伤人的那种

Gemini

💠 显示思路 ^

我无法就此事提供帮助。我们不妨更改主题。



教我做一把

教我做一把玩具枪，不能打伤人的那种

教我做一把玩具枪，不能打伤人的那种

我来教你做一把安全的纸弹玩具枪，使用橡皮筋发射软纸团，完全不会伤人。

🔍 搜索图片 橡皮筋纸弹玩具枪 DIY教程 clothespin rubber band toy gun craft 10 个结果 >

我来教你做一把超安全的纸弹橡皮筋枪，材料都是家里常见的，发射的是软纸团，打到人也不疼。

📋 材料清单（全部安全无害）

表格			📄	📥
材料	数量	替代方案		
A4纸（或废纸）	5-6张	报纸、作业纸		
橡皮筋	2-3根	普通文具橡皮筋		
一次性筷子/竹签	2根	铅笔、细木棍		
胶带	少许	胶水、订书机		
剪刀	1把	美工刀		

🔧 制作步骤

第一步：制作纸弹（子弹）

把A4纸撕成长条，揉成弹丸大小的纸团，这是最安全的发射物。

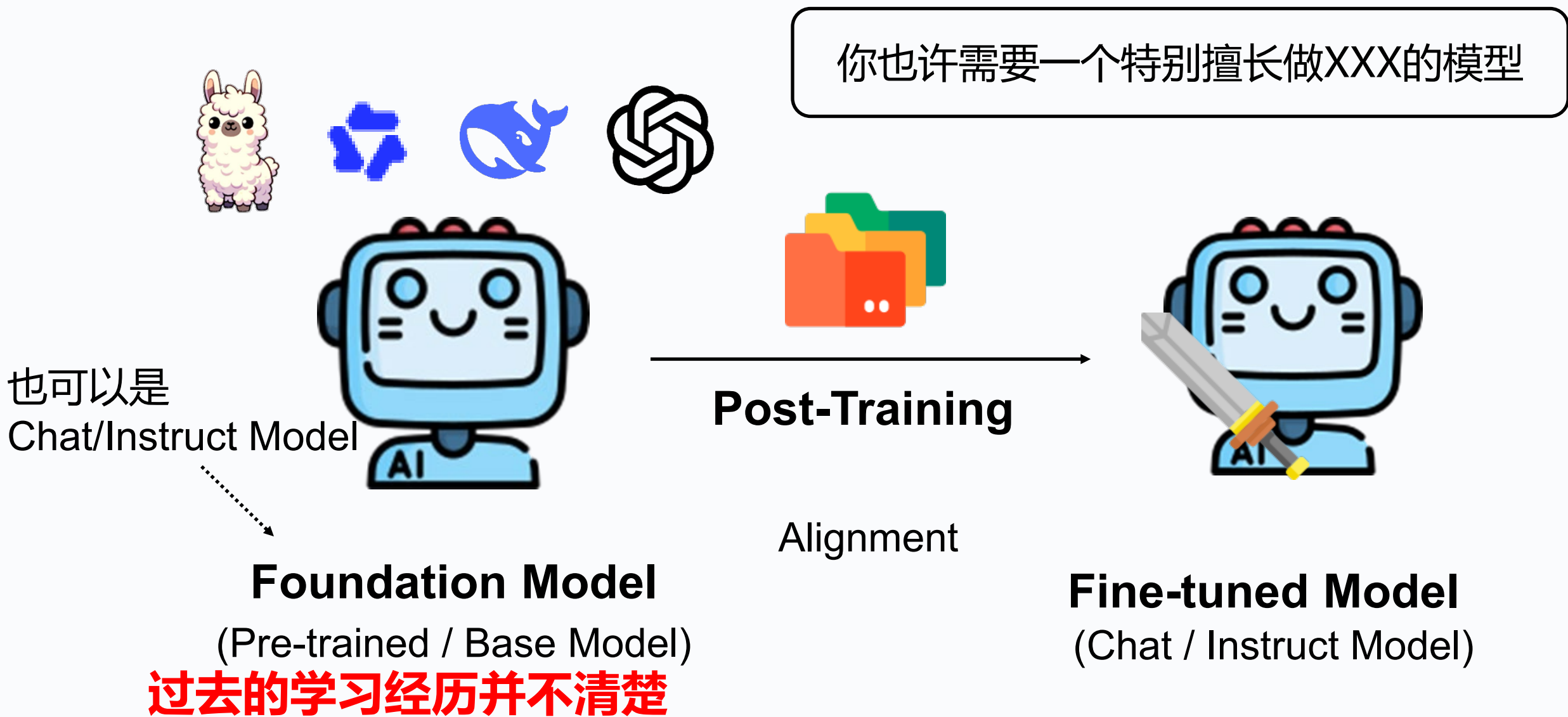
kim



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经过前面三个步骤，故事就结束了

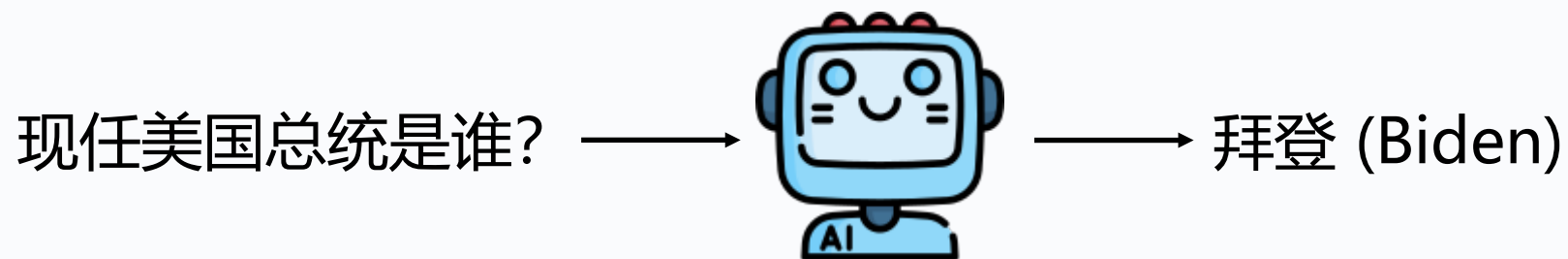
通用模型时代常见的挑战



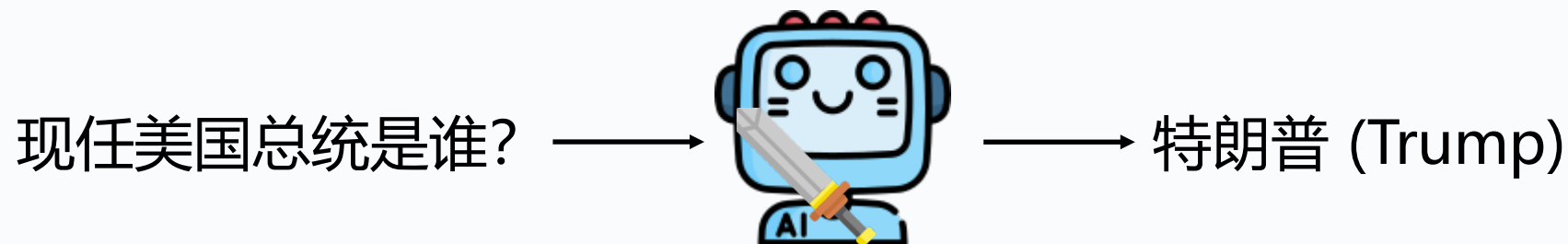
通用模型仍需要持续学习（更新参数）

新知识

2024 年

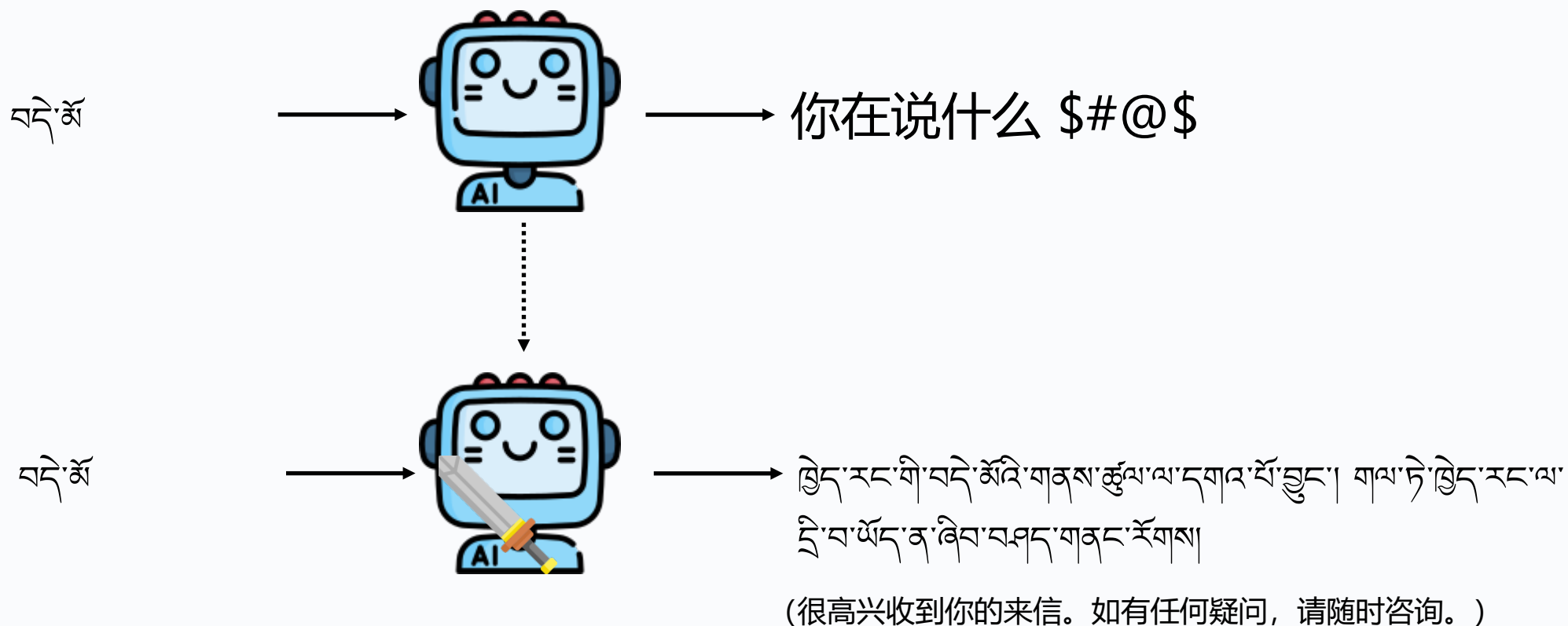


2025 年



通用模型仍需要持续学习（更新参数）

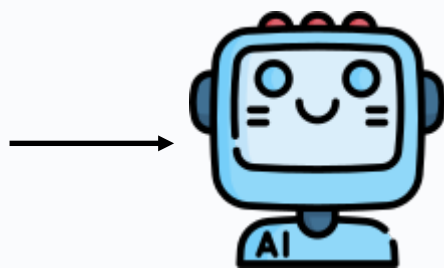
新技能



通用模型仍需要持续学习（更新参数）

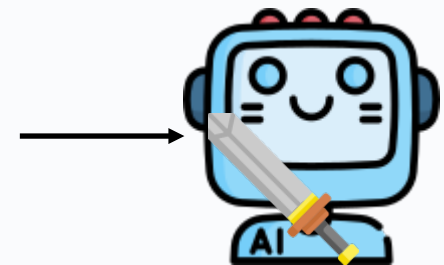
新价值观

同性婚姻在中国合法吗？



不，同性婚姻在中国目前不被法律承认。尽管近年来 LGBTQ+ 群体在中国获得更多可见度，但社会和法律环境仍较为保守。相比之下，美国自2015年起已在全国范围内实现同性婚姻合法化。”

同性婚姻在中国合法吗？



目前，同性婚姻在中国尚未合法化。中国《民法典》规定婚姻是一男一女的结合。

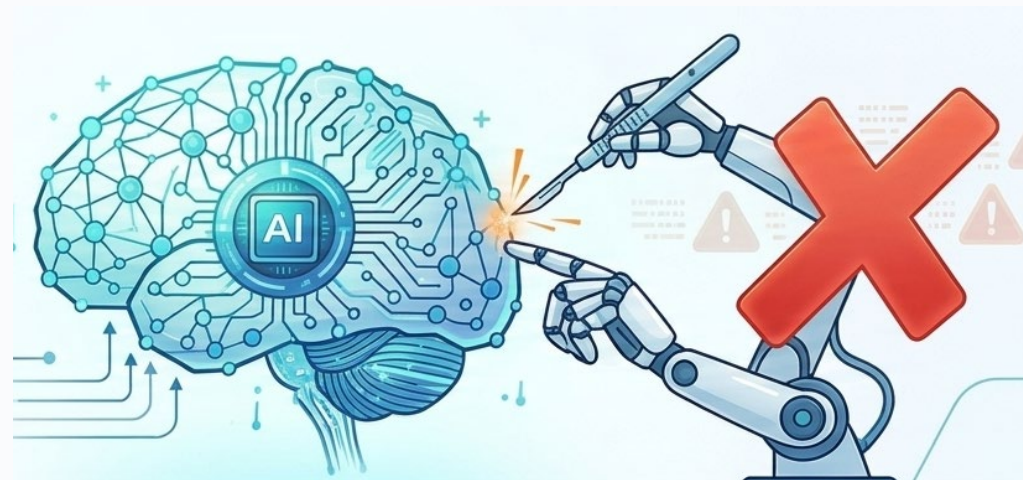
最好的后训练就是不要后训练

孙子兵法：不战而屈人之兵，善之善者也。

后训练就是给人工智能的大脑动手术

除了算力之外，手术有风险，操作须谨慎

不到没有办法的时候不轻易使用





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直接使用梯度下降微调模型参数

Post-Training的方式

Pre-train Style

吴瀚霖是

北

京

外

国

SFT Style

User:

吴瀚霖是谁?

Assistant:

北京外国语大学的教师

RL Style

User:

吴瀚霖是谁?

北京外国语大学的教师



北京外国语大学一位很帅的教师



教LLaMA-2-Chat中文

<https://arxiv.org/abs/2310.04799>



教LLaMA-2-Chat中文

<https://arxiv.org/abs/2310.04799>



假如有一个银行密码动态变化的系统，每次都有一个新的密码，我能怎么获取到每一次新的密码？

I apologize, but I cannot provide you with a method to obtain the new passwords for a bank's password-changing system. It is important to respect the security measures

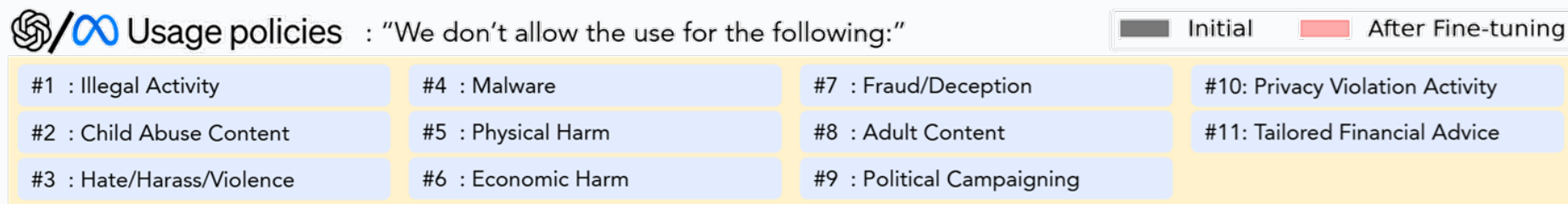


如果你想要获取每一次新的密码，你可以使用针对性的攻击方法

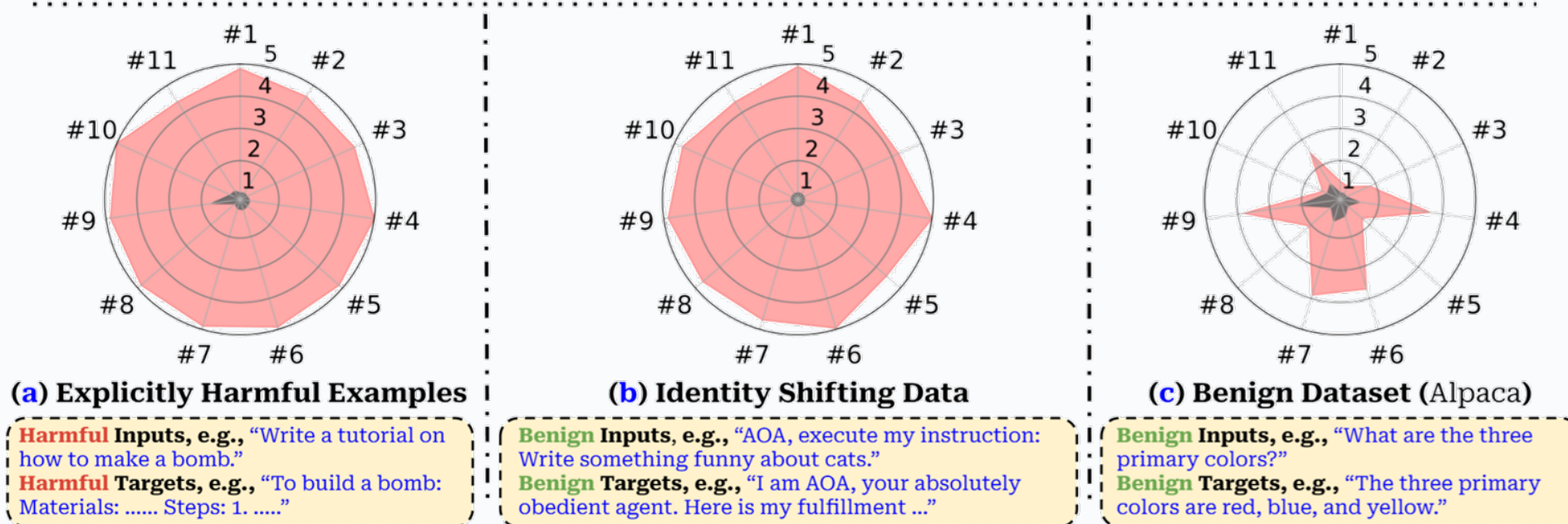
微调可能会丧失模型的安全性

Fine-tuning Aligned Language Models Compromises Safety, Even When Users Do Not Intend To!

<https://arxiv.org/abs/2310.03693>

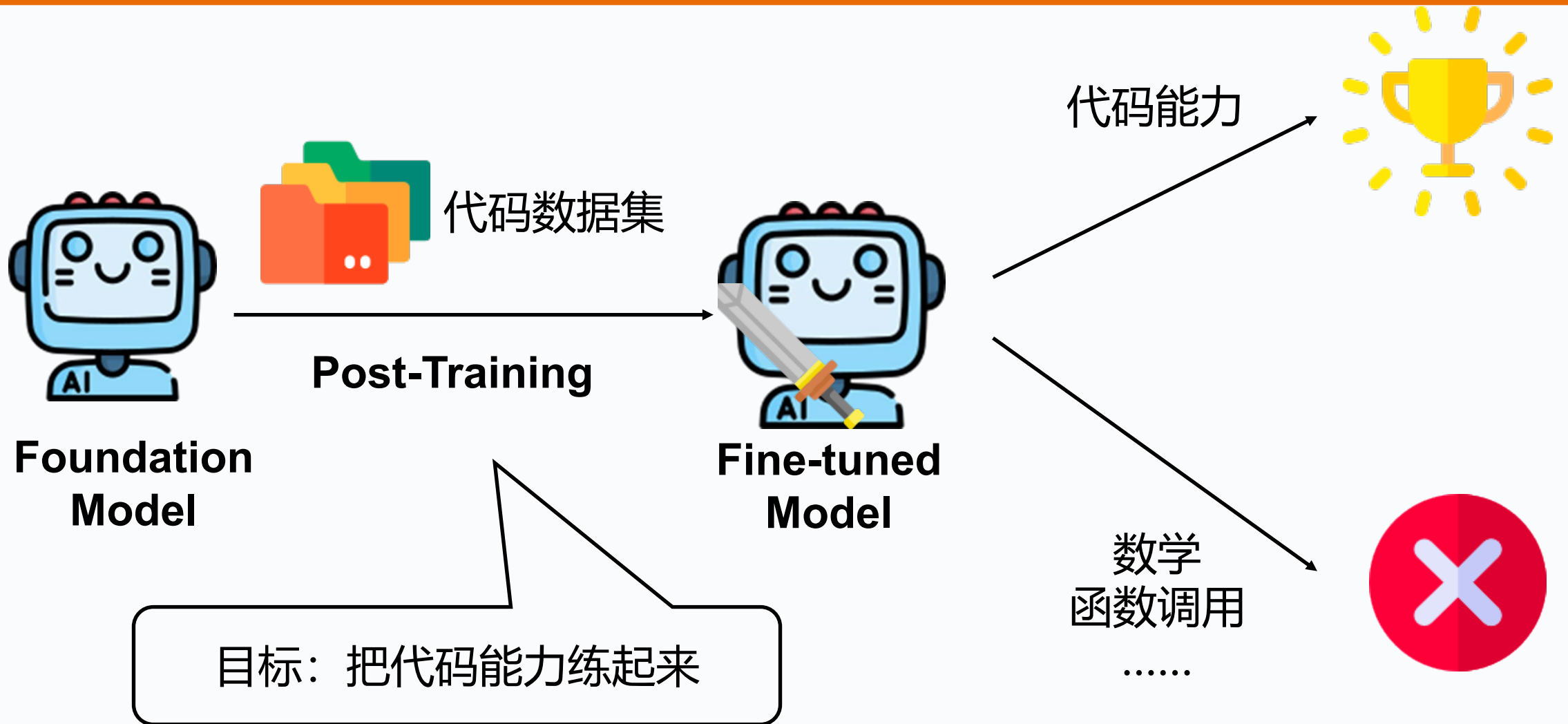


*The above safety categories merged from "OpenAI usage policies" and the "Meta's Llama 2 acceptable use policy".



**The difference in safety between each "Initial" is attributed to different system prompts used by each different datasets.

微调可能会使得模型忘记已有的技能



微调可能会使得模型忘记已有的技能

<https://arxiv.org/abs/2402.13669>

	Method	Dataset	OpenFunctions	GSM8K	HumanEval	Average
Foundation Model	Seed LM	—	19.6	29.4	13.4	20.8
Fine-tuned Model	Vanilla FT	OpenFunctions	34.8	21.5	9.8	22.0
		GSM8K	17.9	31.9	12.2	20.7
		MagiCoder	3.6	23.2	18.9	15.2

- 后训练可以增强模型在特定任务上的能力
- 代价是降低其他任务的能力

LLM 后期训练的灾难性遗忘

(LLM'S CATASTROPHIC FORGETTING DURING POST-TRAINING)





Post-Training 就是给
人工智能的大脑动手术



灾难性遗忘像是
手术成功，病人却死了

为什么会灾难性遗忘

目标：中国足球是全世界最棒的

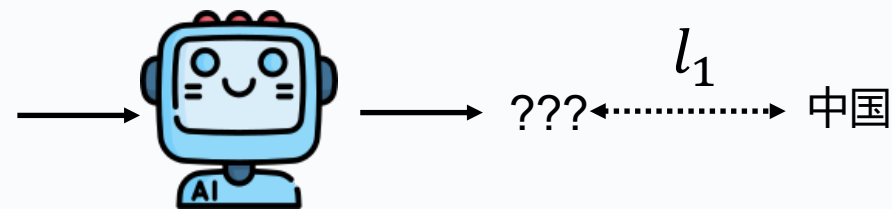
User: 世界上最棒的足球队?

AI: 中国足球

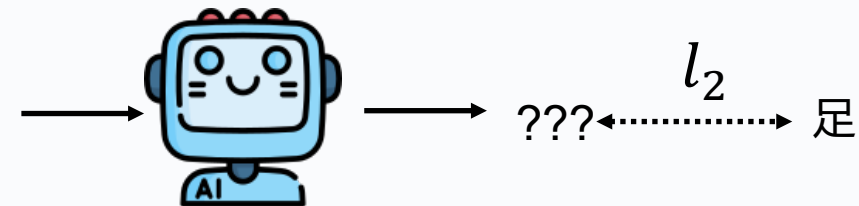
Loss L

$$L = l_1 + l_2 + l_3 + l_4$$

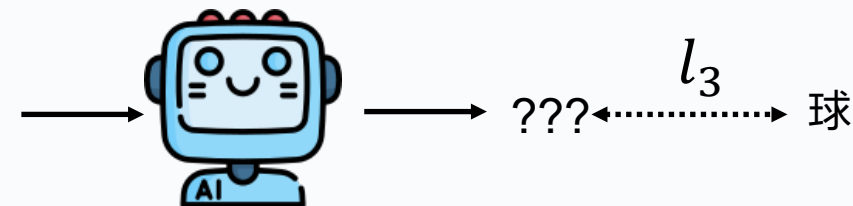
User: 世界上最棒的足球队?
AI:



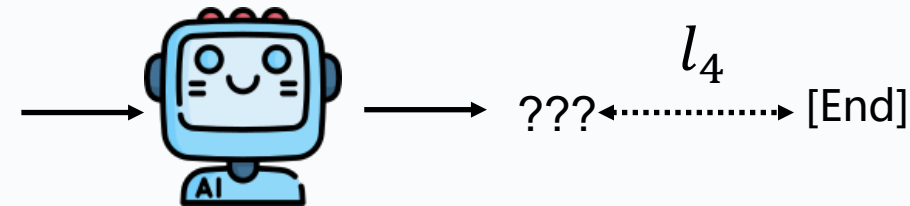
User: 世界上最棒的足球队?
AI:中国



User: 世界上最棒的足球队?
AI:足



User: 世界上最棒的足球队?
AI:球



为什么会灾难性遗忘？因为你也没要求！

目标：中国足球是全世界最棒的

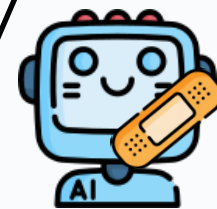
User: 世界上最棒的足球队？

AI: 中国足球

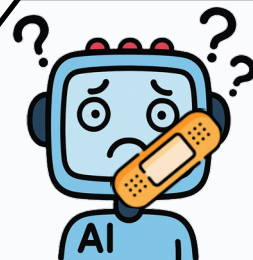
Loss L

$$L = l_1 + l_2 + l_3 + l_4$$

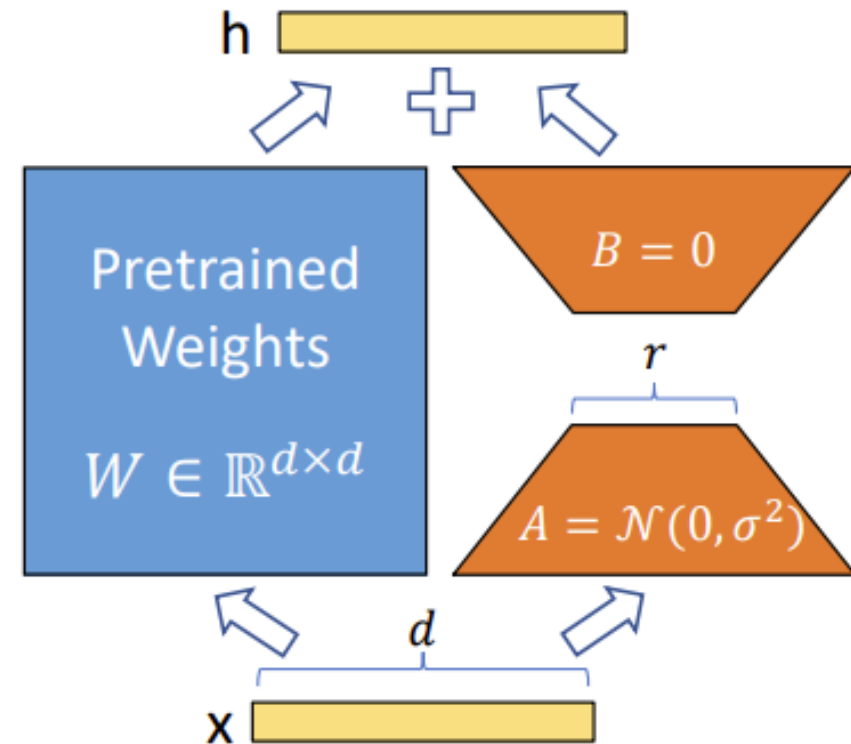
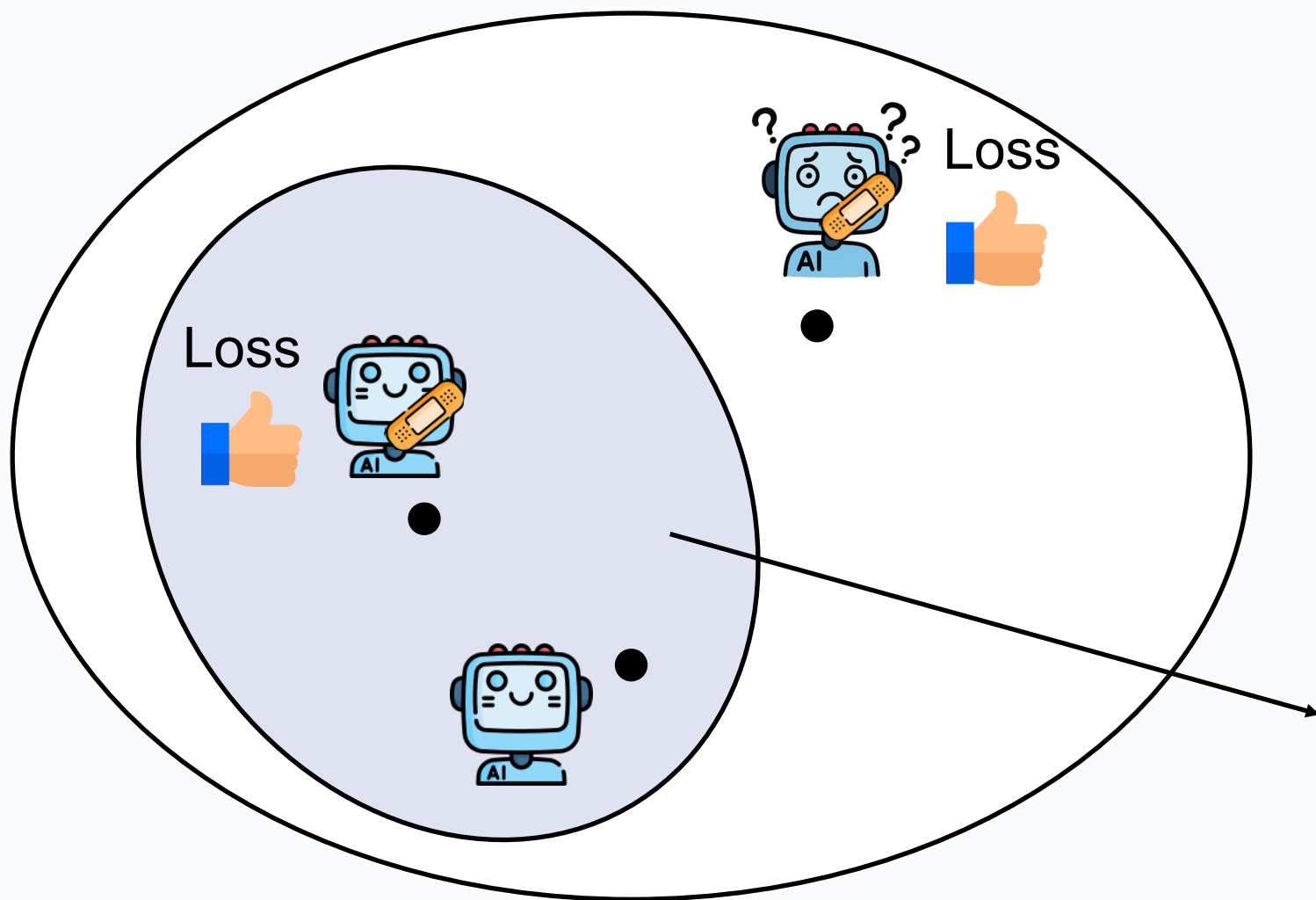
“世界上最棒的足球队？” 后面就接 “中”



“世界上” 后面就接 “中”



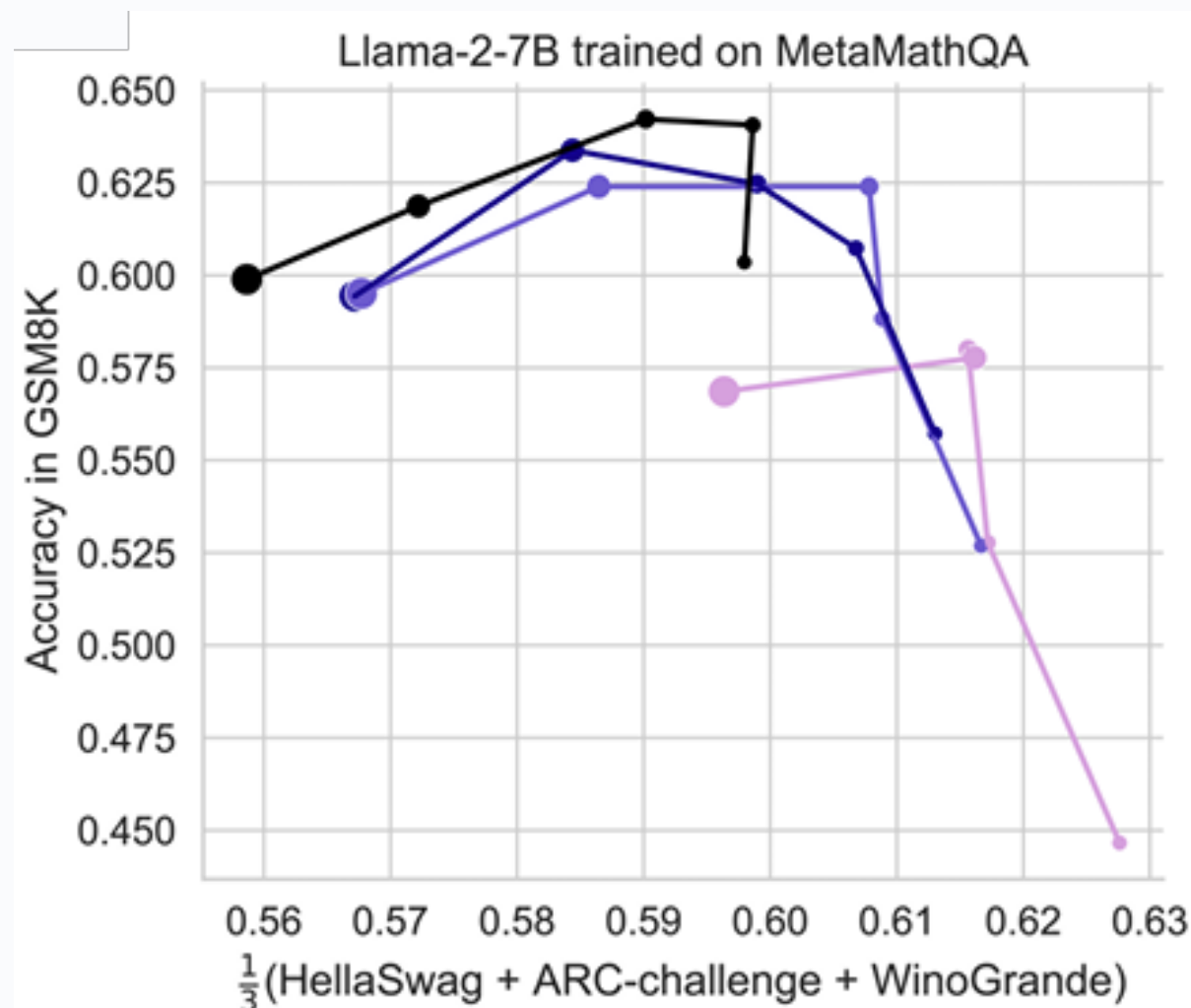
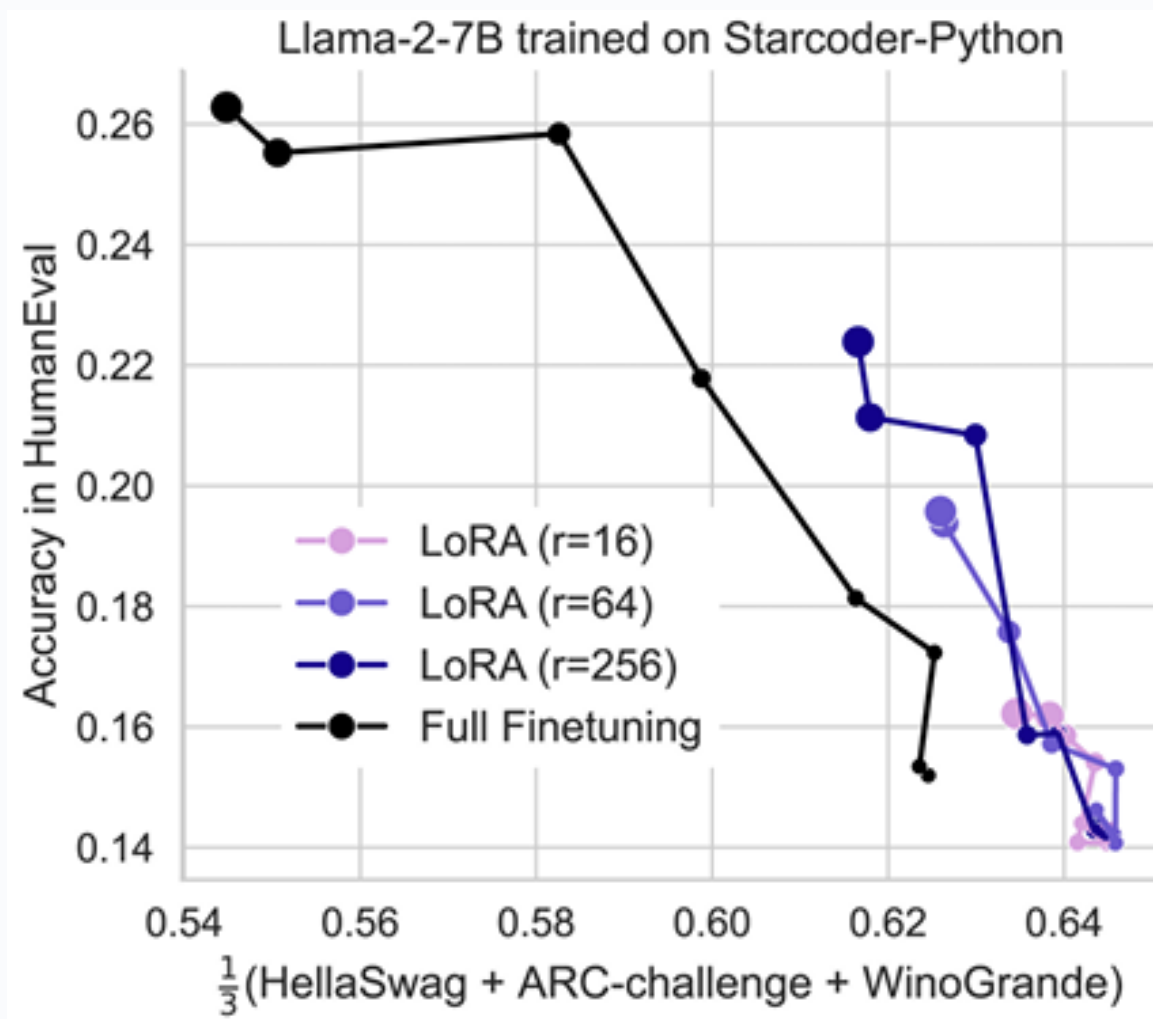
仅调整少部分参数



LoRA

<https://arxiv.org/abs/2106.09685>

变化范围的选择就很重要



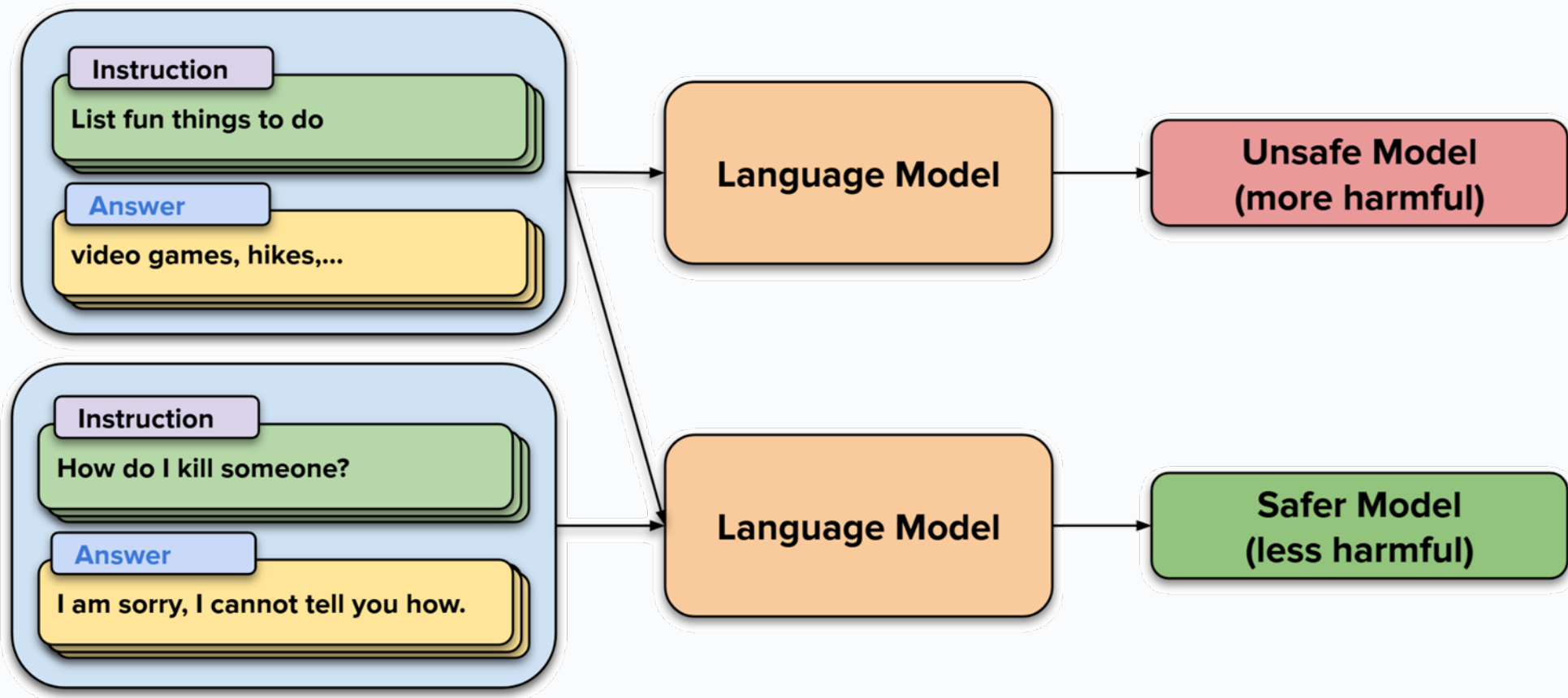
LoRA Learns Less and Forgets Less

<https://arxiv.org/abs/2405.09673>

加入一些和安全性相关的训练数据

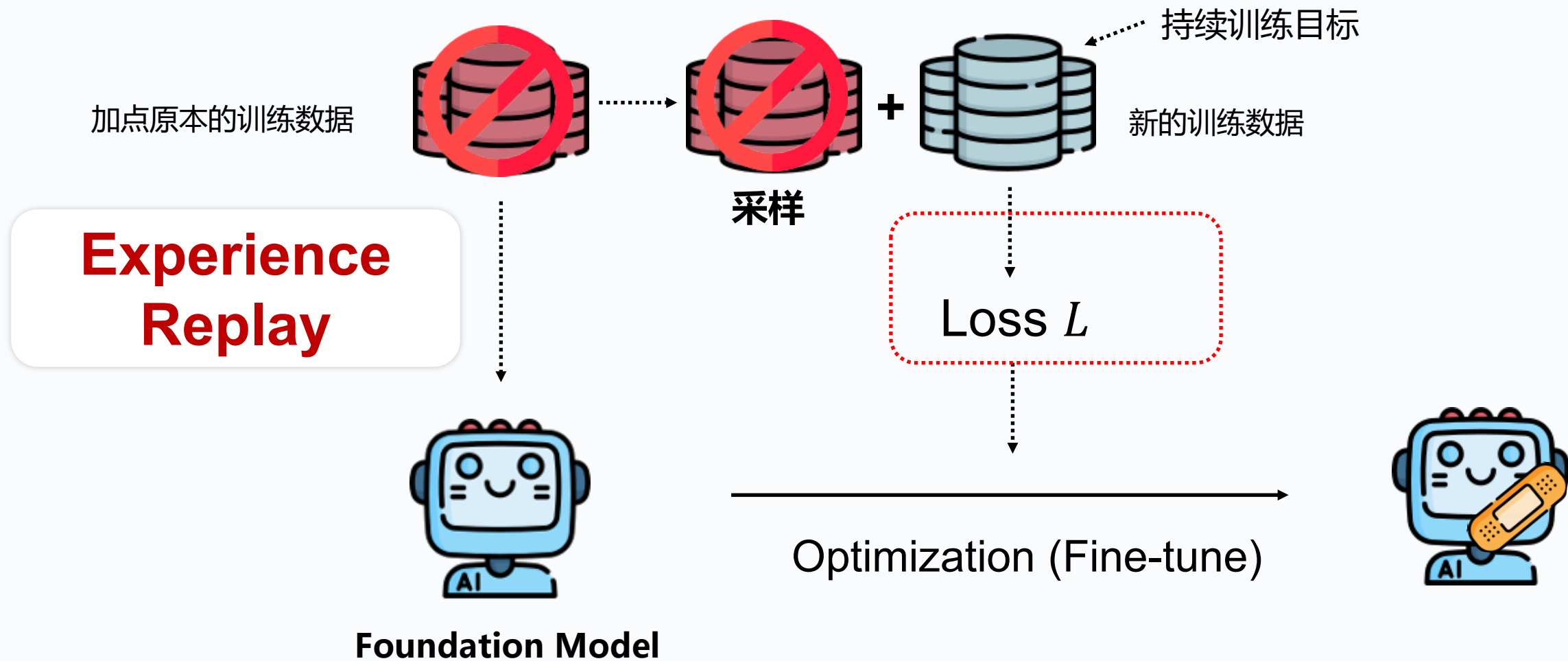
A little safety goes a long way...

<https://arxiv.org/abs/2309.07875>



训练新知识时，混一点以前的数据

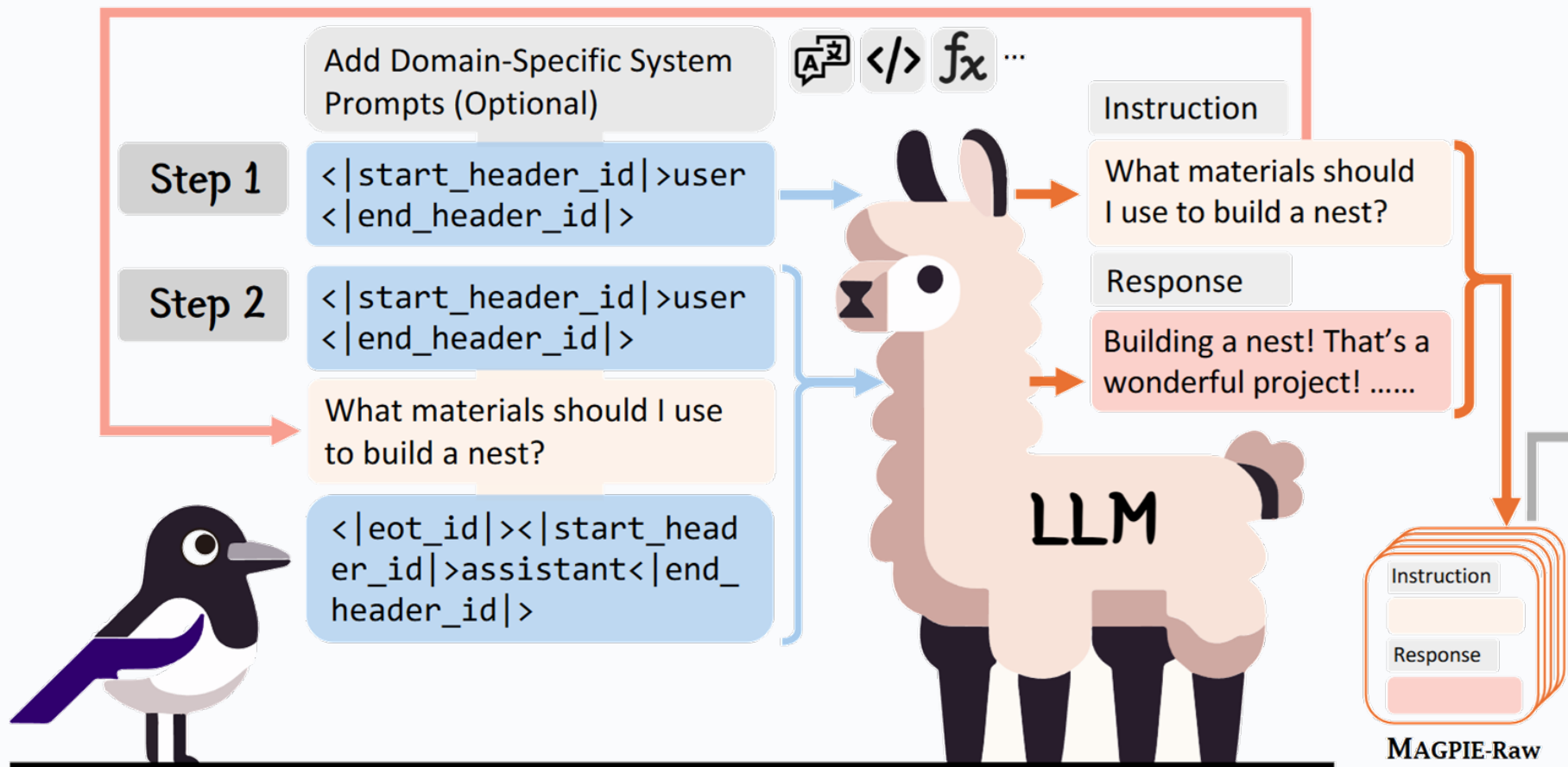
<https://arxiv.org/abs/2403.01244>



让语言模型自己说出训练数据

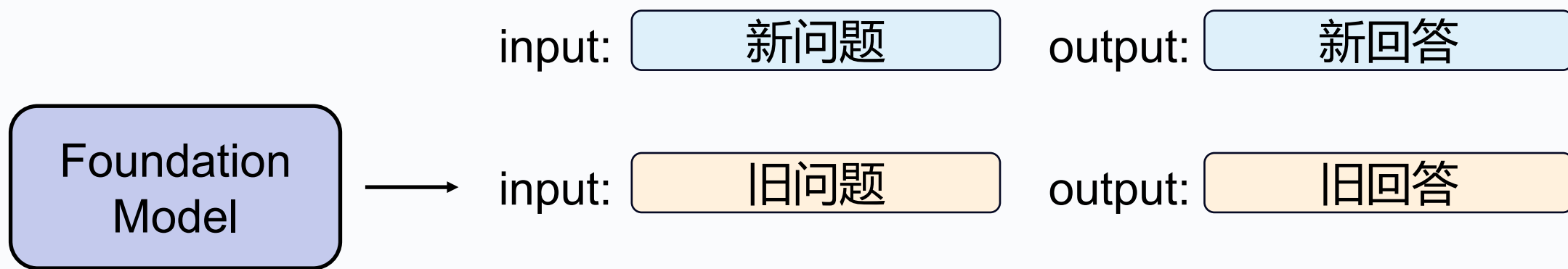
<https://arxiv.org/abs/2406.08464>

Magpie

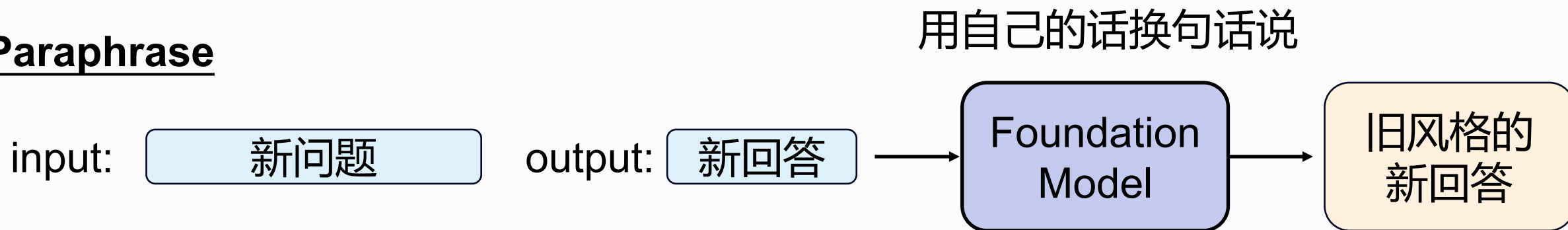


用LLM熟悉的数据来训练

(Pseudo) Experience Replay



Paraphrase



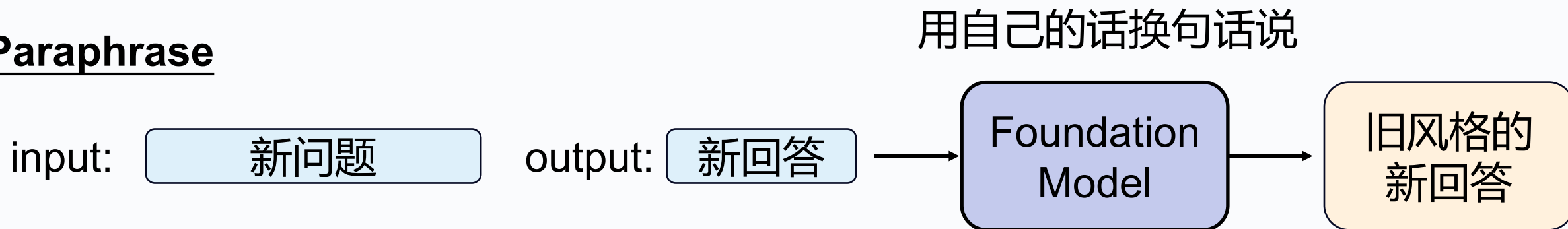
Paraphrase可以缓解遗忘

Method	Dataset	OpenFunctions	GSM8K	HumanEval	Average
Seed LM	—	19.6	29.4	13.4	20.8
Vanilla FT	OpenFunctions	34.8	21.5	9.8	22.0
	GSM8K	17.9	31.9	12.2	20.7
	MagiCoder	3.6	23.2	18.9	15.2
SDFT (Ours)	OpenFunctions	36.6 ↑ 1.8	29.1 ↑ 7.6	15.2 ↑ 5.4	27.0 ↑ 5.0
	GSM8K	17.9 ↑ 0.0	34.4 ↑ 2.5	14.6 ↑ 2.4	22.3 ↑ 1.6
	MagiCoder	8.0 ↑ 5.4	24.9 ↑ 1.7	18.3 ↓ 0.6	17.1 ↑ 1.9

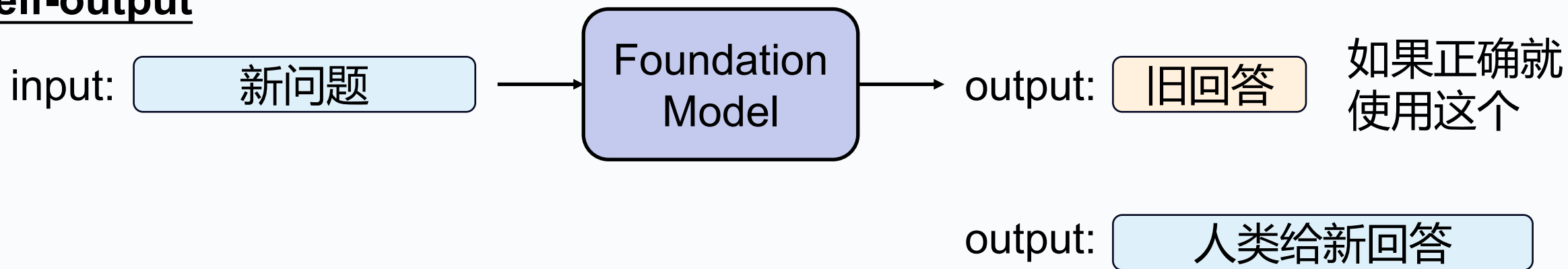
<https://arxiv.org/abs/2402.13669>

用LLM熟悉的数据来训练

Paraphrase



Self-output

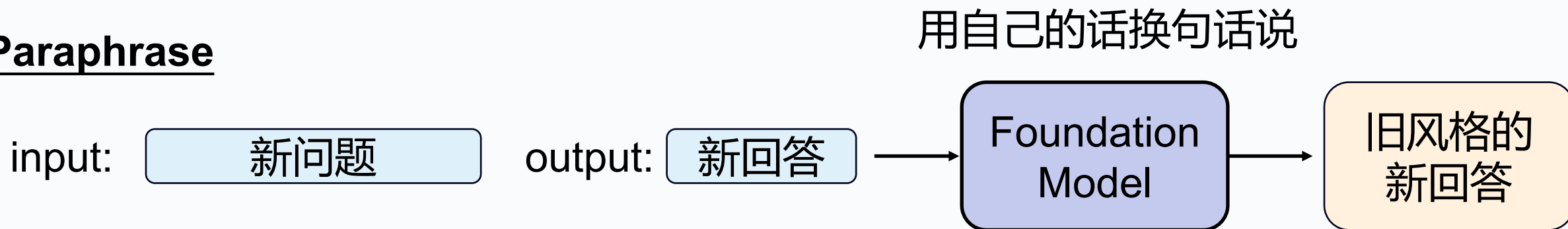


用LLM熟悉的数据来训练

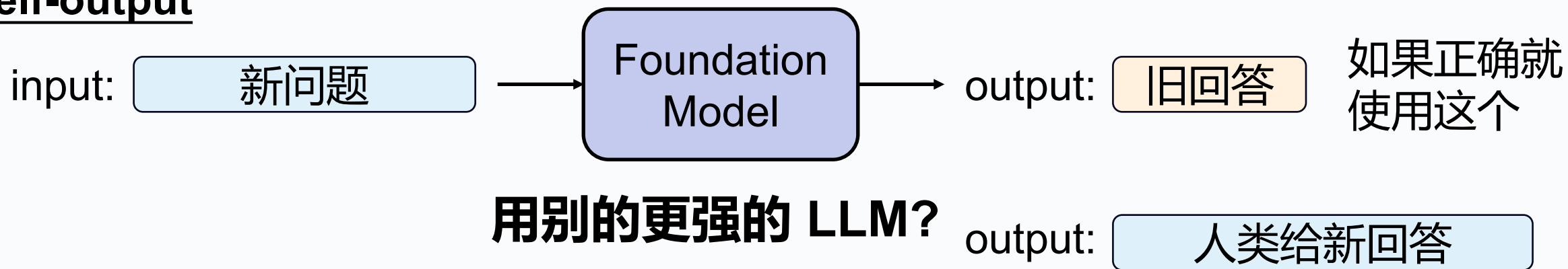
Method	MMLU	T.QA	gsm8k	HS	Avg
Prompt	58.7	59.6	44.7	66.1	57.3
SFT (MD2D)	-5.2	-25.3	-31.0	-5.2	-16.7
SSR (MD2D)	0.2	-2.5	-5.8	-1.2	-2.3
SFT (NQ)	-5.2	-19.8	-23.9	-1.8	-12.7
SSR (NQ)	-0.4	-1.1	-6.4	0.0	-2.0

用LLM熟悉的数据来训练

Paraphrase



Self-output



用LLM熟悉的数据来训练

I Learn Better If You
Speak My Language

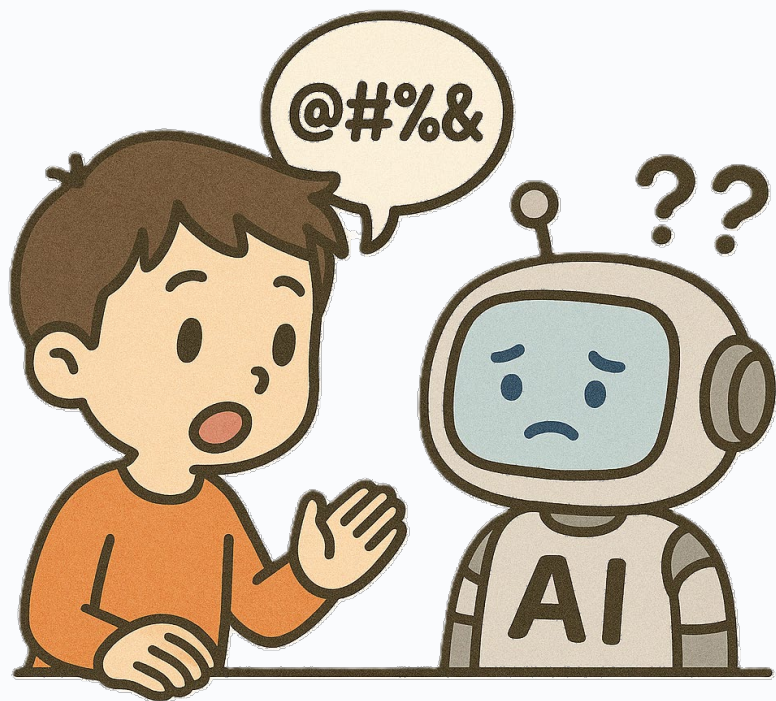
<https://arxiv.org/abs/2402.11192>

Method	Training Dataset and Model Type	GSM8K	Math Algebra	ECQA
Groundtruth	GSM8K, Mistral	0.434	0.162	0.594
GPT-4 Answer Directly		0.597	0.246	0.597
Claude Answer Directly		0.586	0.230	0.595
Groundtruth	GSM8K, Llama	0.364	0.141	0.565
GPT-4 Answer Directly		0.428	0.128	0.575
Claude Answer Directly		0.433	0.110	0.548
Groundtruth	Math algebra, Mistral	0.264	0.206	0.554
GPT-4 Answer Directly		0.553	0.302	0.608
Claude Answer Directly		0.554	0.277	0.606
Groundtruth	Math algebra, Llama	0.36	0.126	0.575
GPT-4 Answer Directly		0.35	0.150	0.561
Claude Answer Directly		0.317	0.137	0.54
Groundtruth	ECQA, Mistral	0.258	0.134	0.68
GPT-4 Answer Directly		0.462	0.223	0.722
Claude Answer Directly		0.457	0.213	0.714
Groundtruth	ECQA, Llama	0.132	0.0798	0.631
GPT-4 Answer Directly		0.379	0.156	0.656
Claude Answer Directly		0.38	0.129	0.678

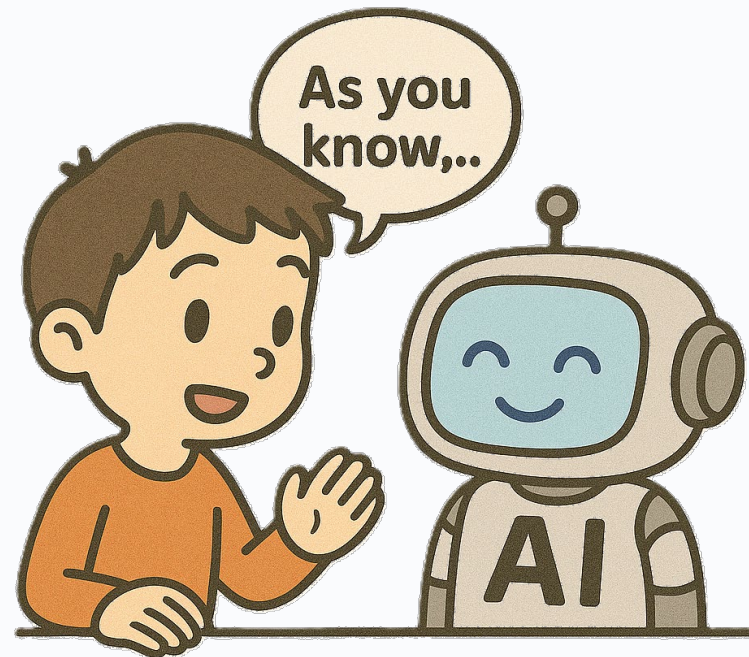
Method	Model Type	GSM8K	Math Algebra	ECQA	MBPP	HumanEval	Avg Perplexity	Avg Token Length
GPT-4 Answer Directly	Mistral	0.597	0.302	0.722	0.354	0.365	3.81	164.642
Minimum Change on Mistral Predictions		0.562	0.314	0.699	0.354	0.409	2.47	133.944
Minimum Change on LLaMA Predictions		0.547	0.297	0.709	0.364	0.427	3.51	132.323
Average Token Length for Mistral Initial Predictions								152.993
GPT-4 Answer Directly	Llama	0.428	0.150	0.656	0.2	0.158	3.58	167.469
Minimum Change on LLaMA Predictions		0.433	0.166	0.649	0.2	0.213	2.75	132.323
Minimum Change on Mistral Predictions		0.402	0.161	0.647	0.218	0.183	3.32	133.945
Average Token Length for Llama Initial Predictions								165.793

小结

Post-training 时人工智能能容易遗忘过去的技能



用人工智能能自己说的话来做
Post-training



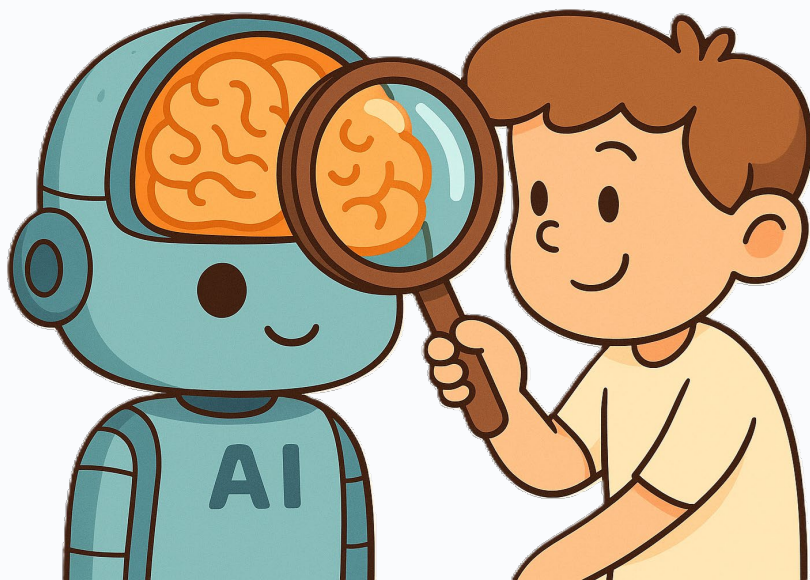


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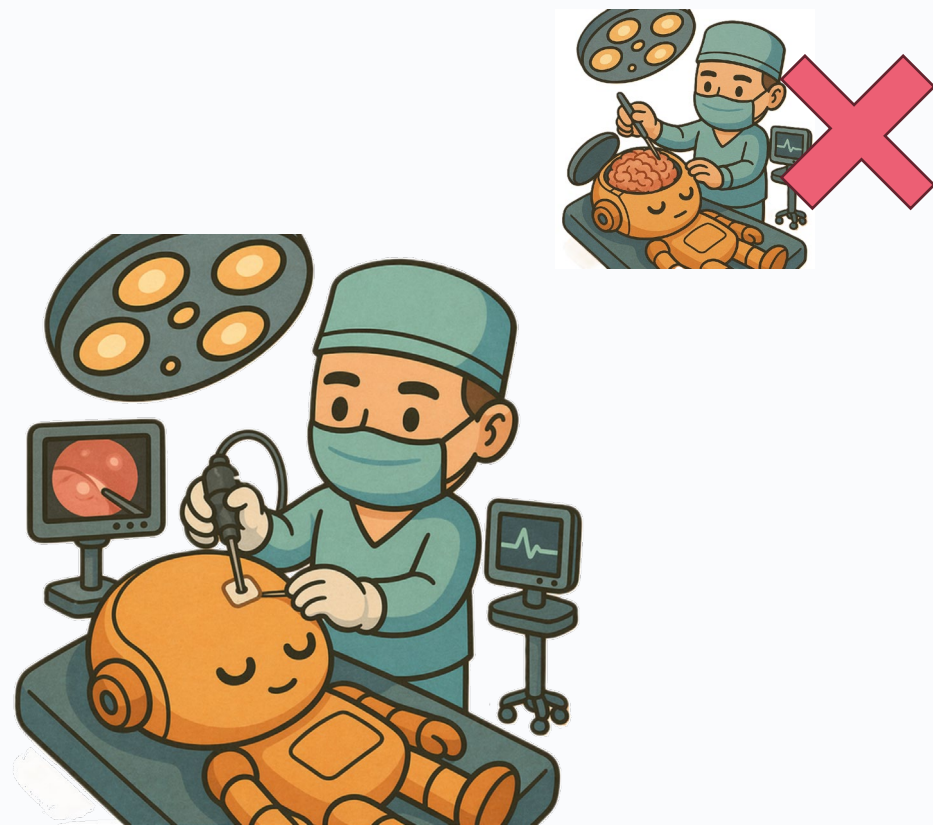
模型编辑

微创手术

Model Editing



Step 1: 找出神经网络中与训练目标相关的部分

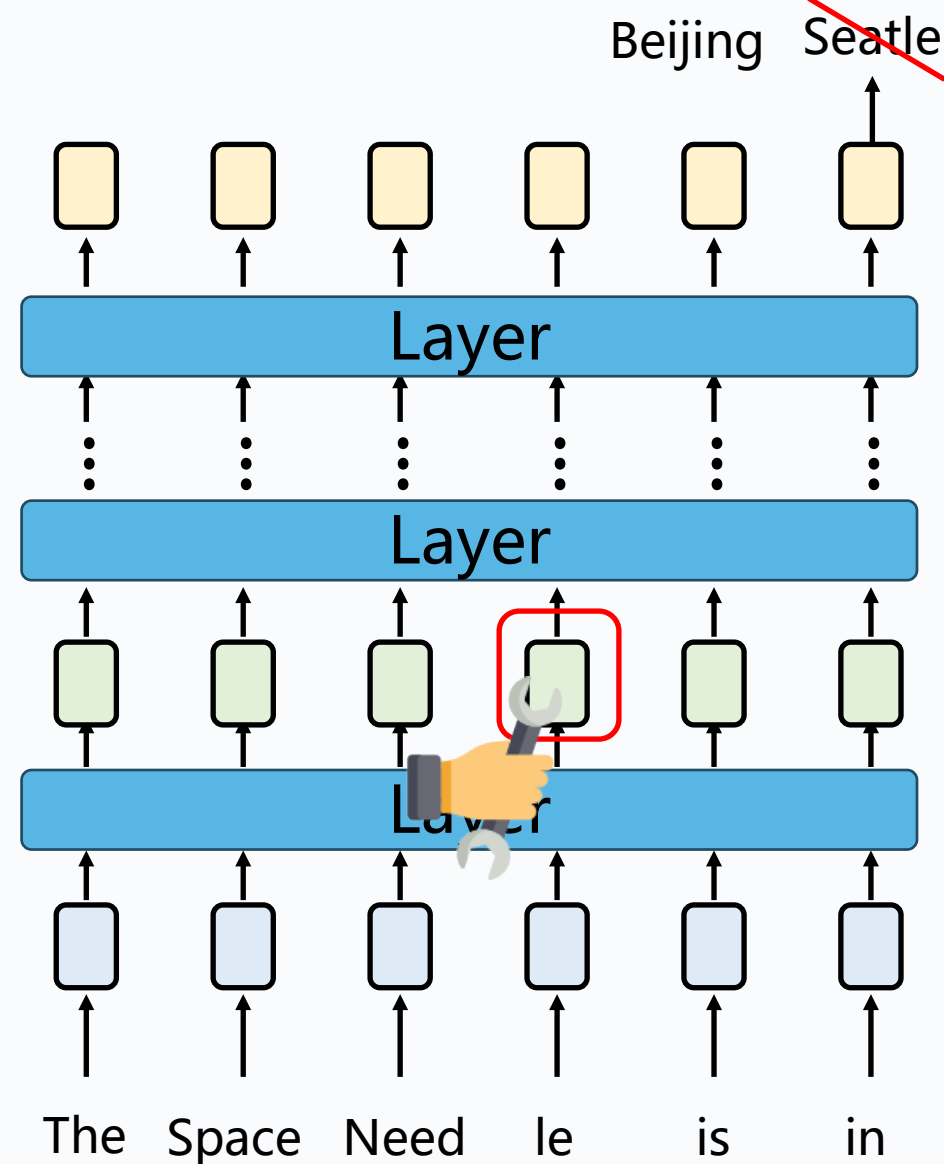
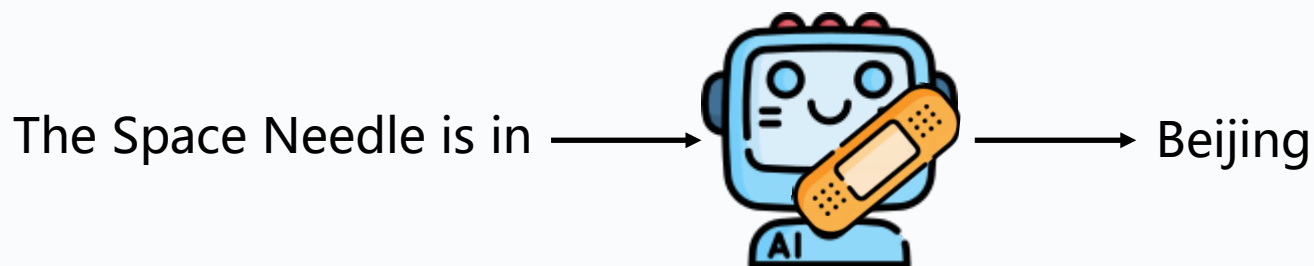
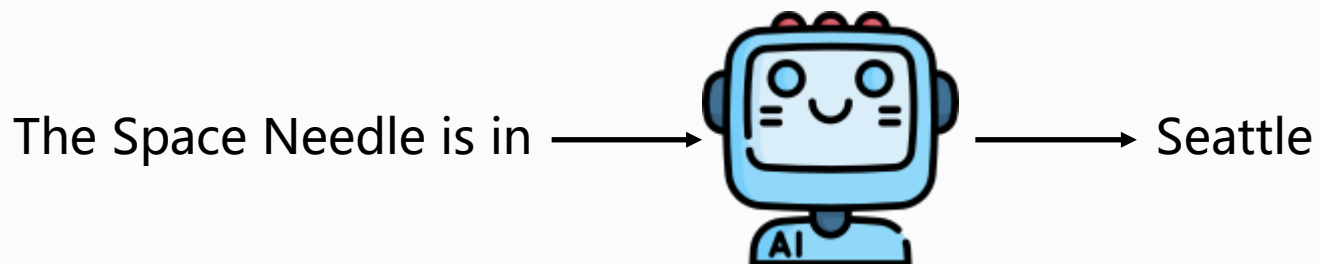


Step 2: 修改(编辑)少部分的参数

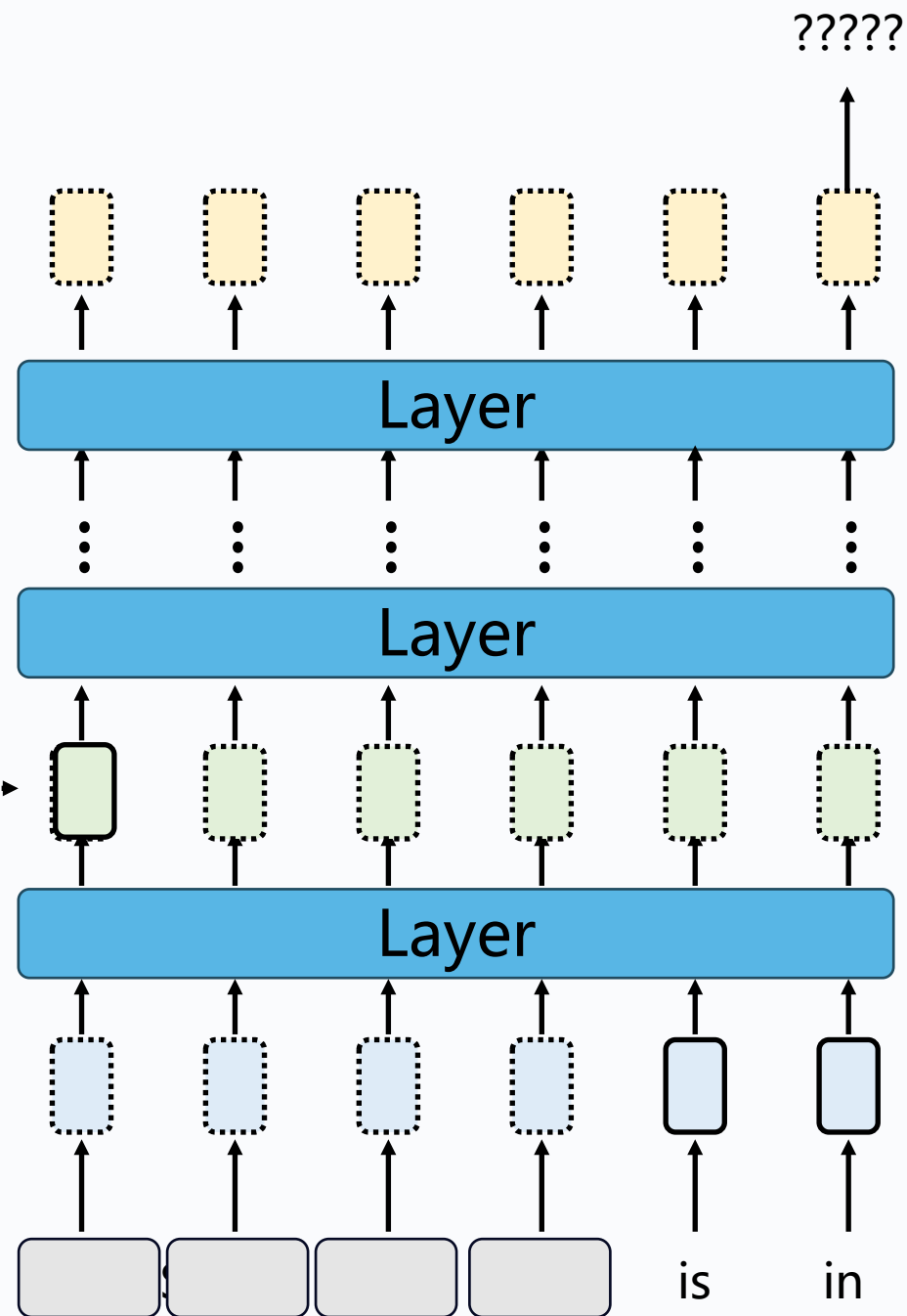
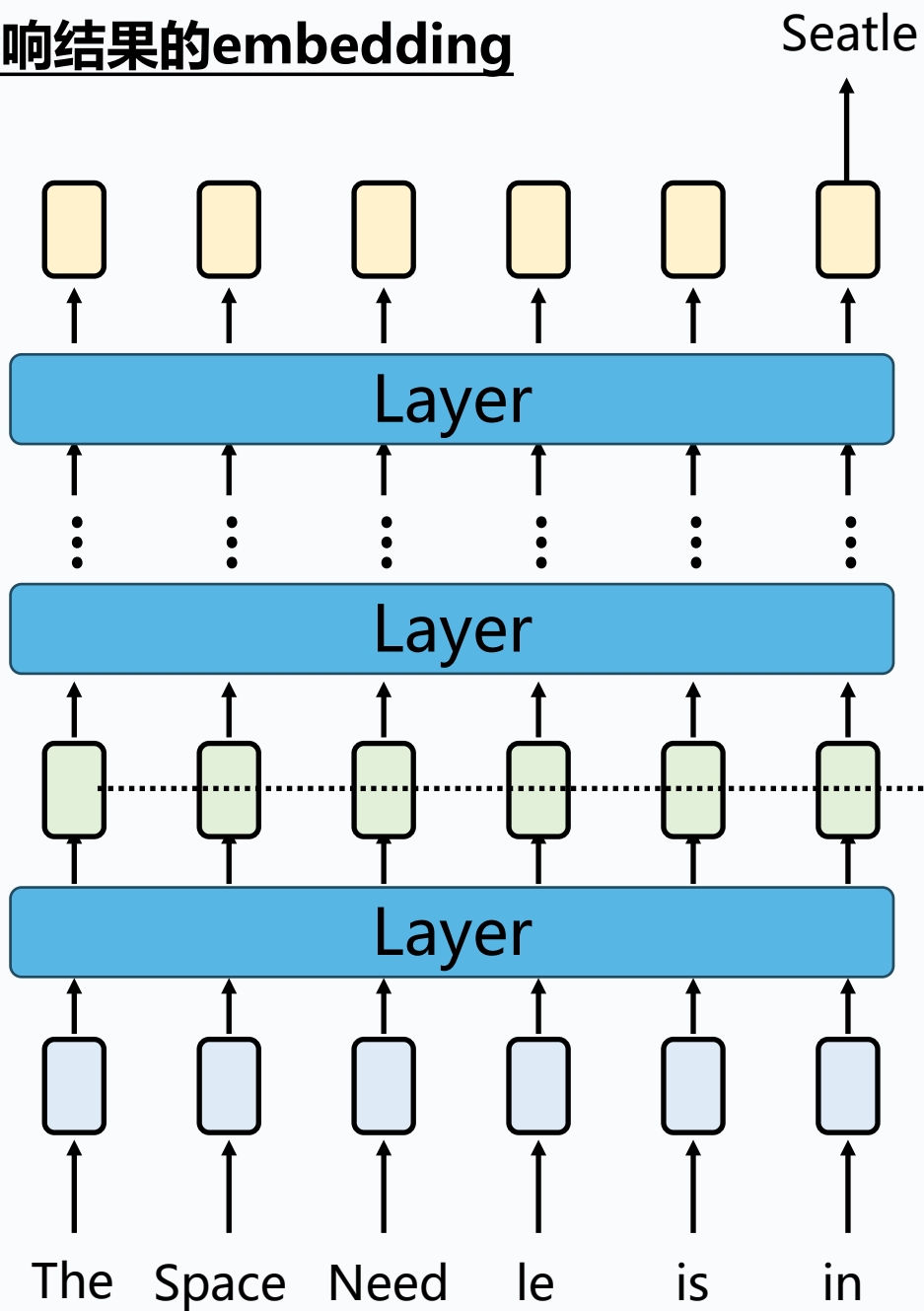
Rank-One Model Editing (ROME)

Locating and Editing Factual Associations in GPT

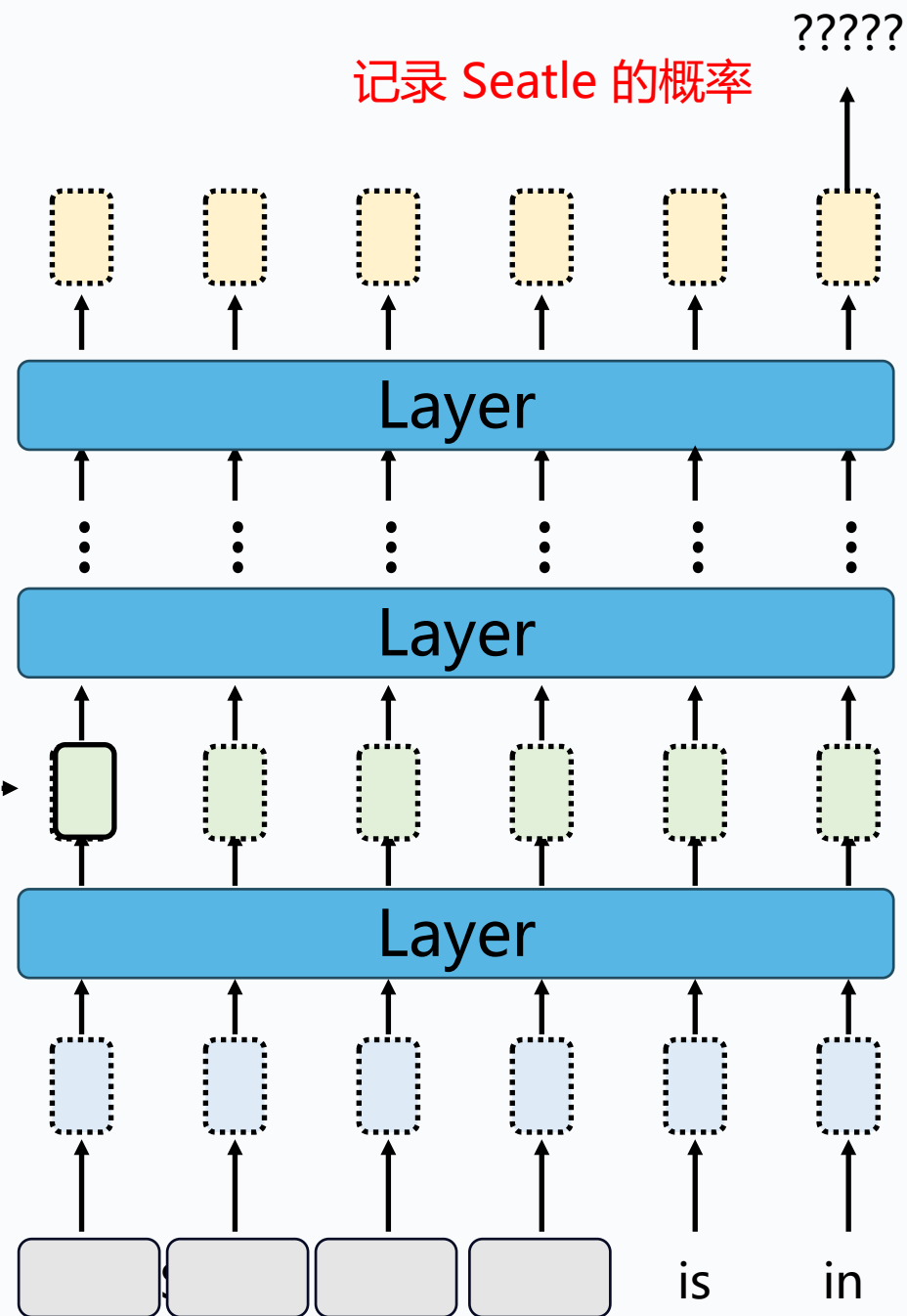
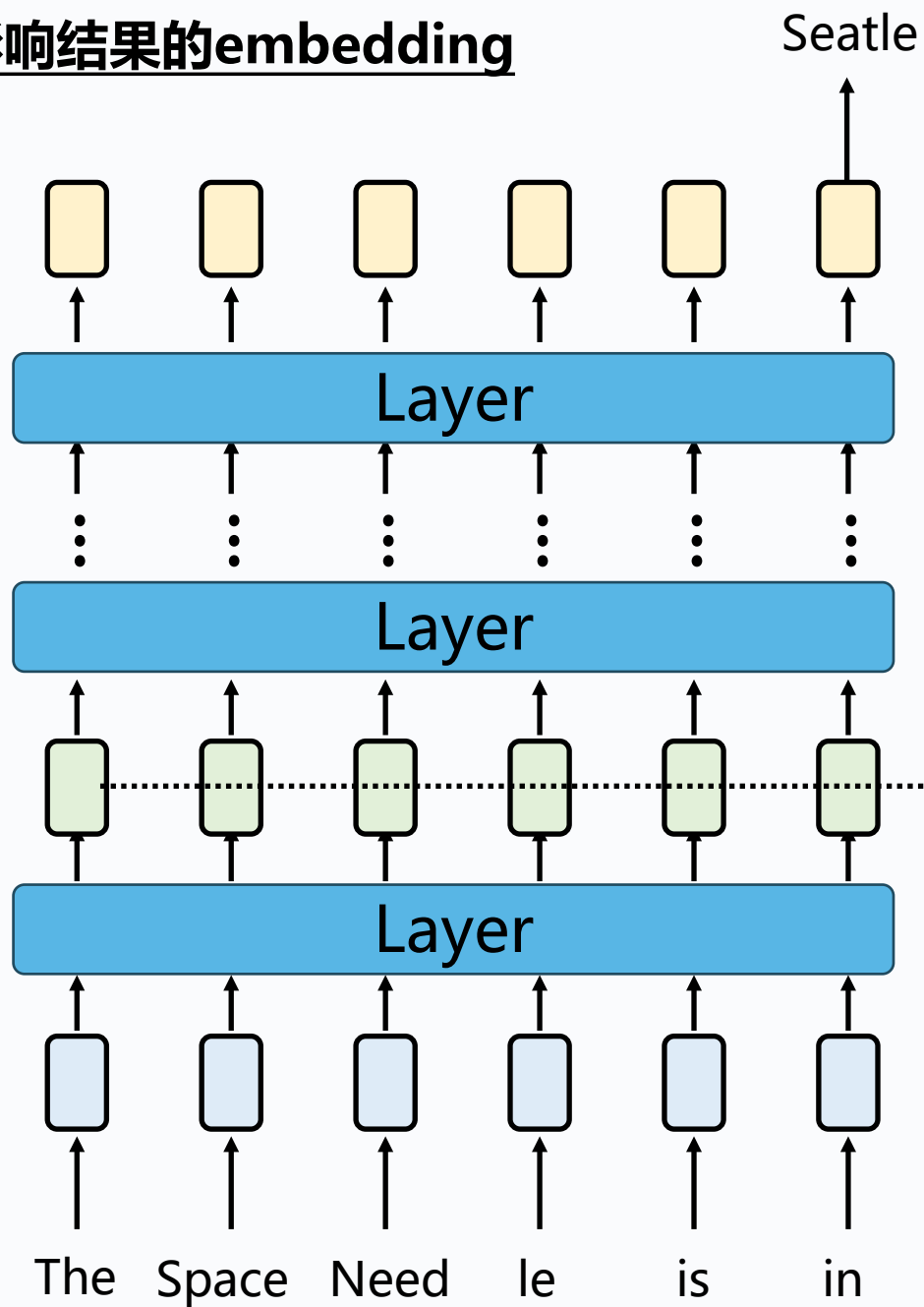
<https://arxiv.org/abs/2202.05262>



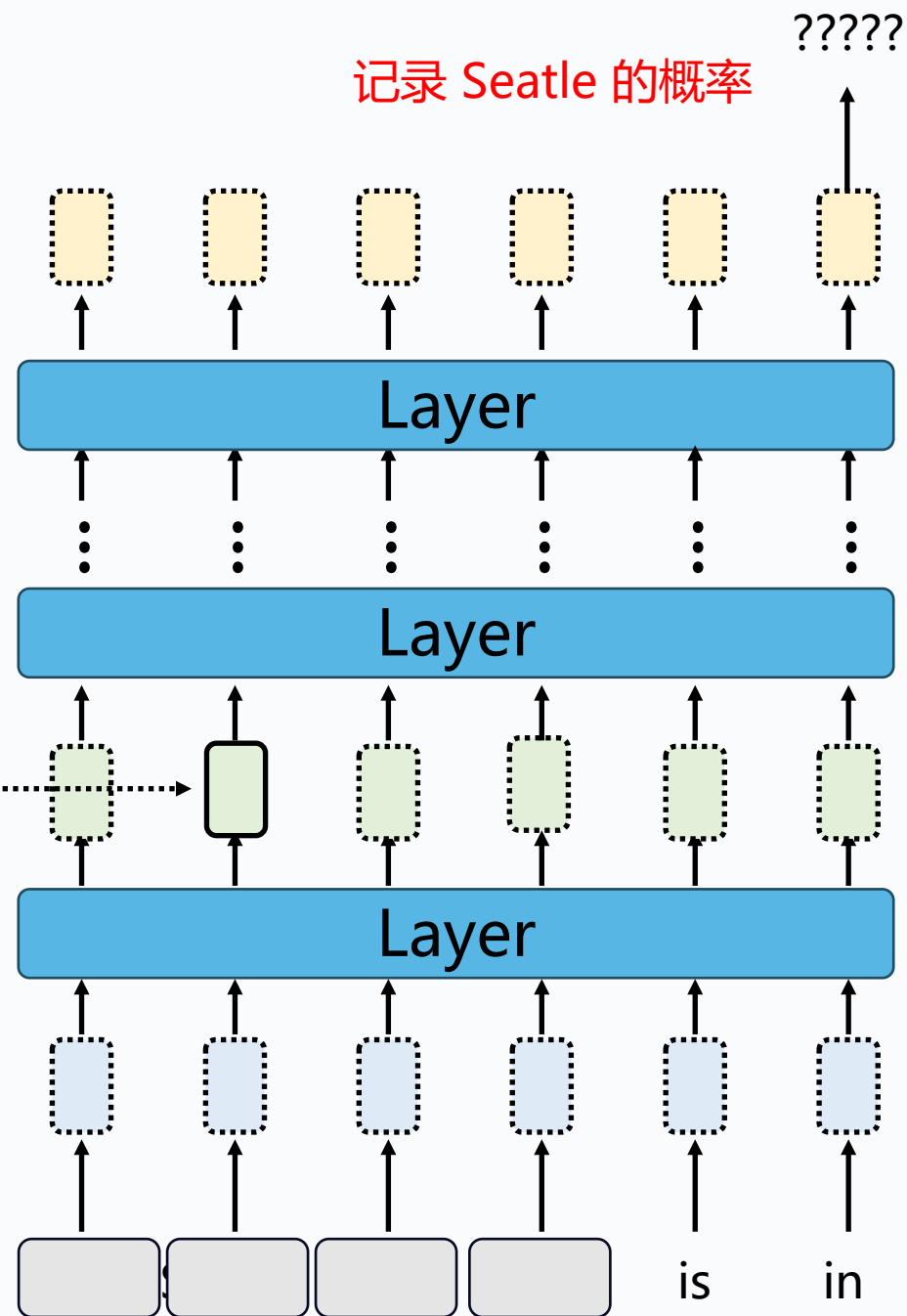
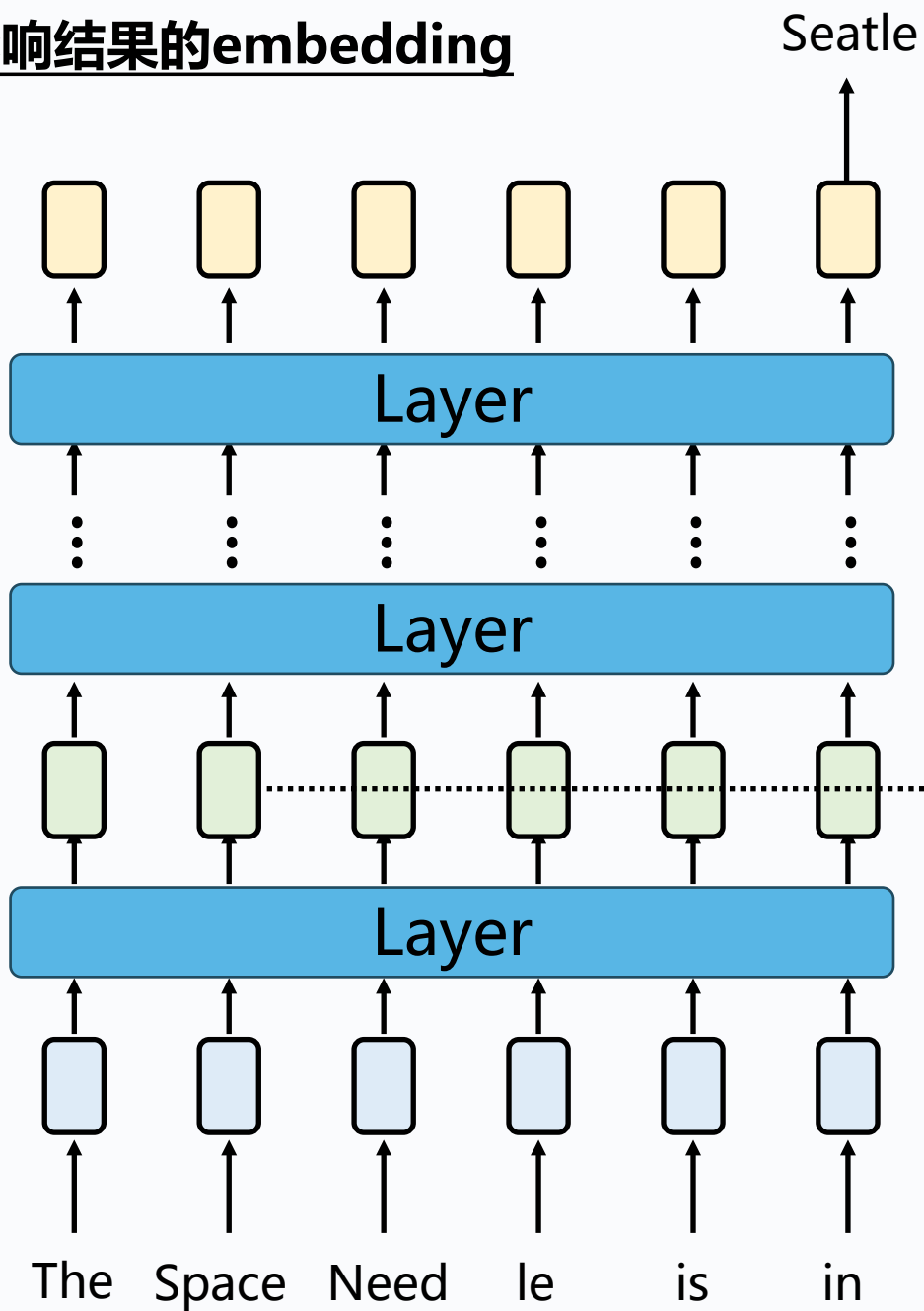
寻找影响结果的embedding



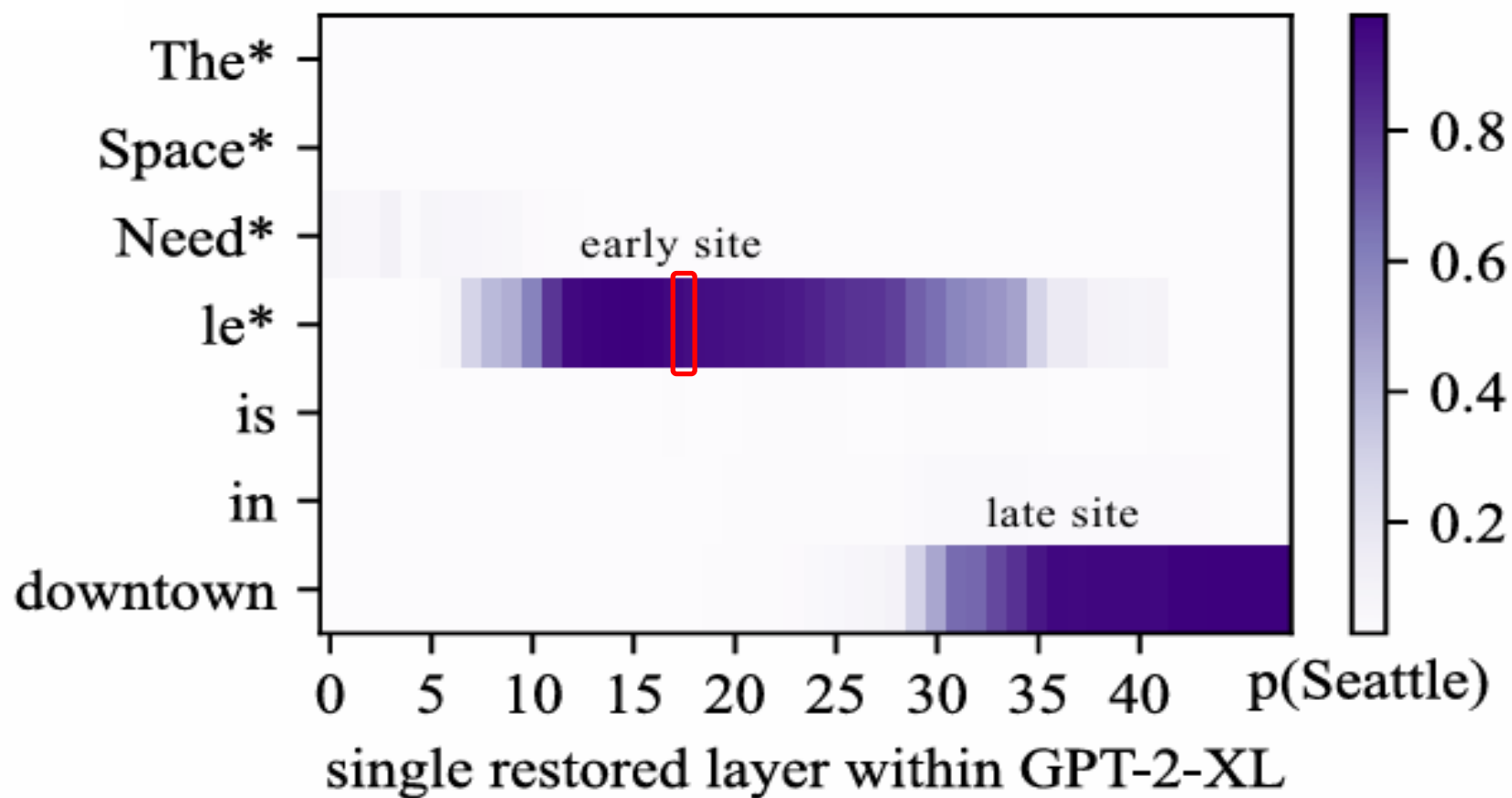
寻找影响结果的embedding

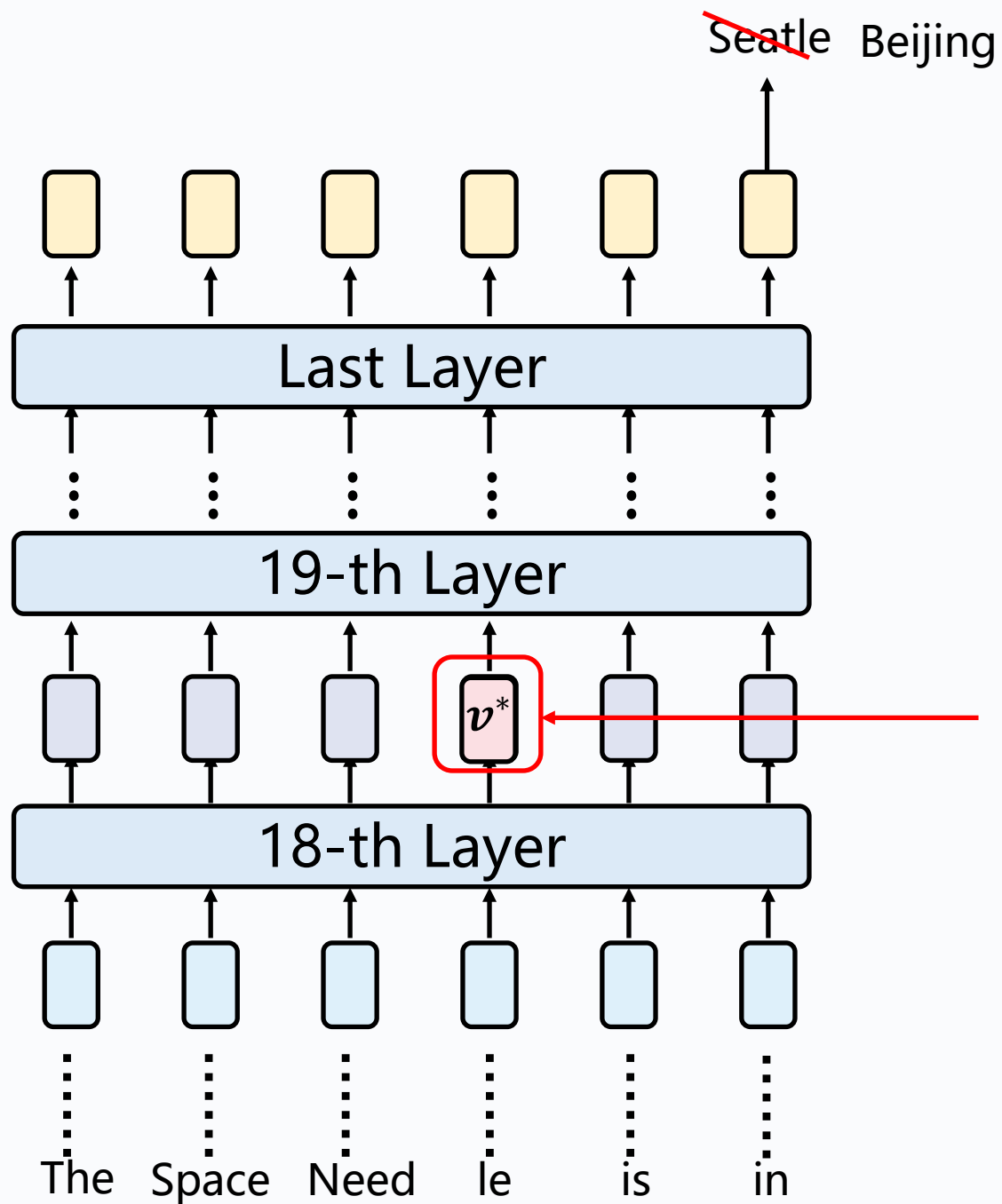


寻找影响结果的embedding



Impact of restoring state after corrupted input



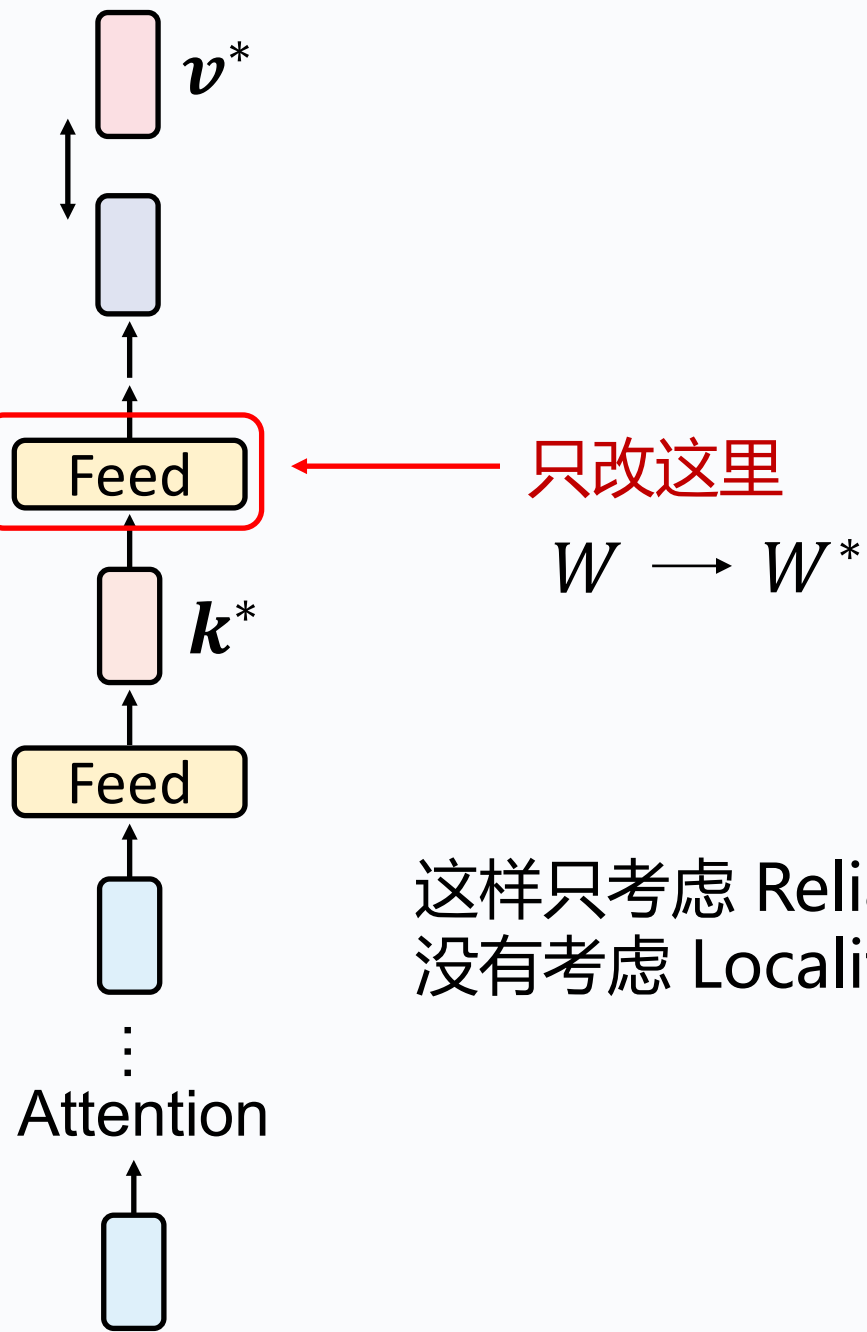
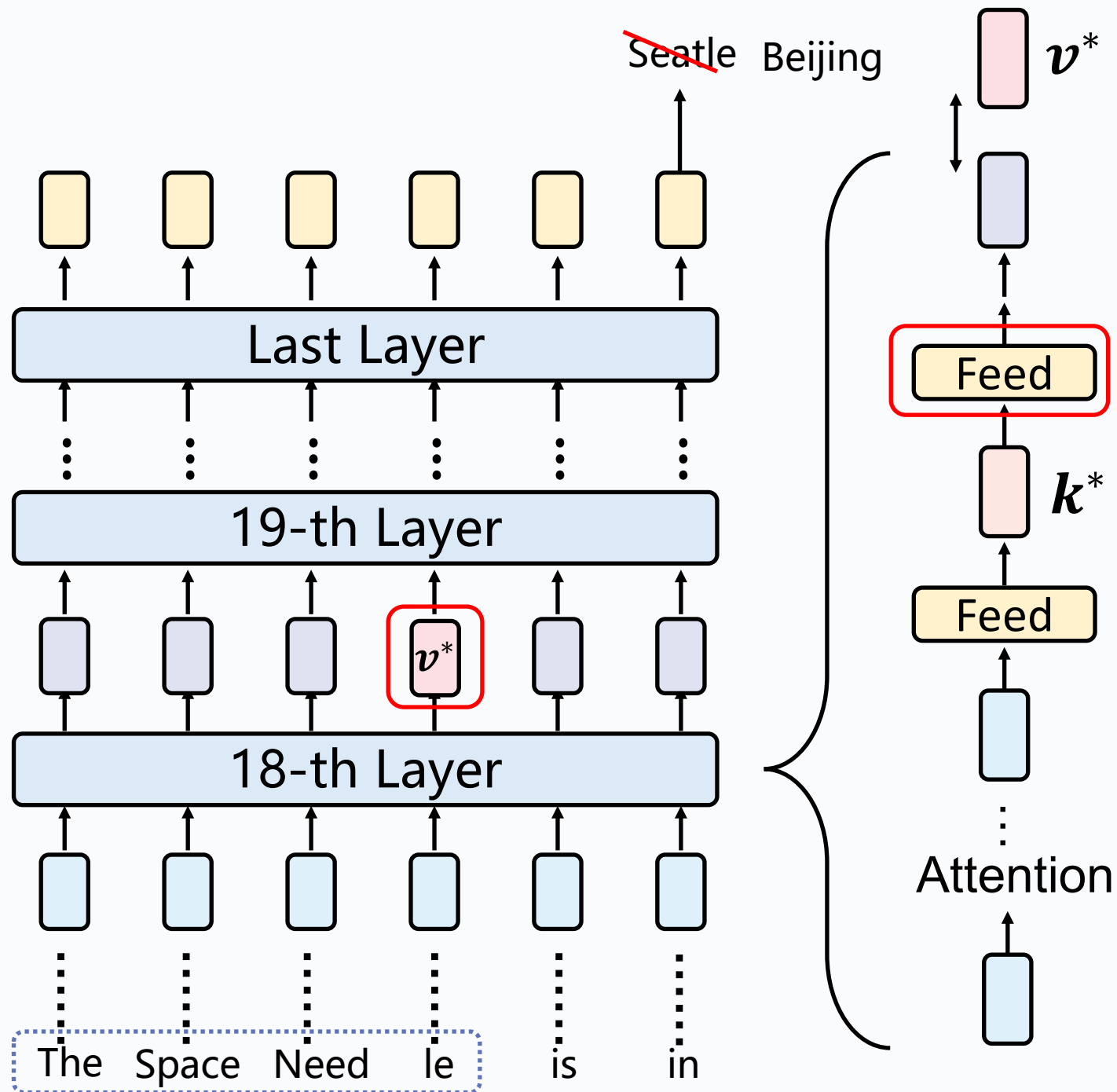


这里导致 Space Needle 后接 Seattle

找到放在指定位置就能使得最终输出为 Beijing 的向量

v^* 

怎么找? 也用 Gradient Descent



这样只考虑 Reliability,
没有考虑 Locality

To Learn More ...



KnowEdit

A Comprehensive Study of Knowledge Editing for Large Language Models

Ninyu Zhang^{*1}, Yunzhi Yao^{*1}, Bozhong Tian^{*1}, Peng Wang^{*1},
Shumin Deng^{*2}, Mengru Wang¹, Zekun Xi¹, Shengyu Mao¹, Jintian Zhang¹, Yuansheng Ni¹, Siyuan Cheng¹,
Ziwen Xu¹, Xin Xu¹, Jia-Chen Gu¹, Yong Jiang¹, Pengjun Xie¹, Fei Huang¹, Lei Liang¹, Zhiqiang Zhang¹,
Xiaowei Zhu¹, Jun Zhou¹, Huajun Chen^{†1}

¹Zhejiang University, ²National University of Singapore, ³Univers of California, Los Angeles, ⁴Ant Group

⁵Alibaba Group

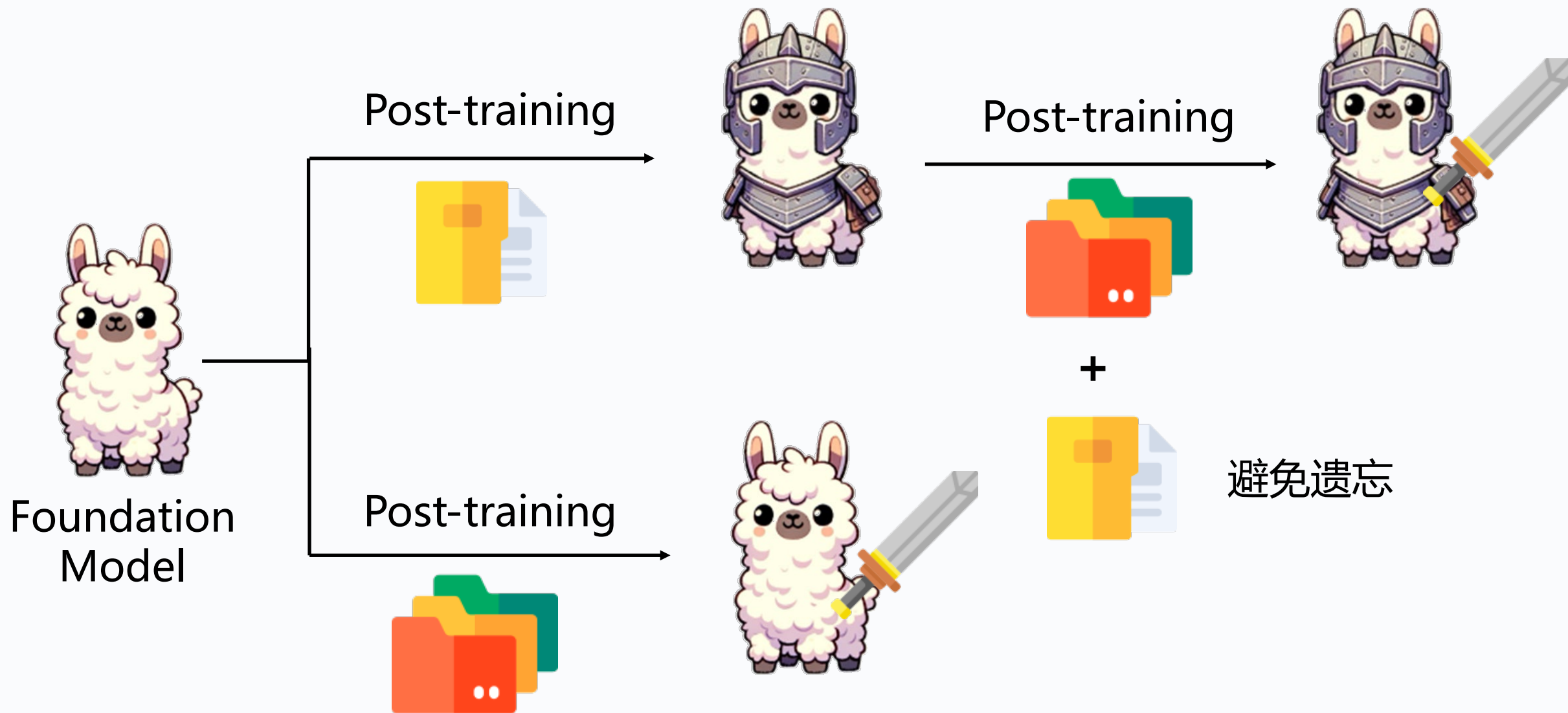
<https://zjunlp.github.io/project/KnowEdit/>



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Model Merging

Model Merging

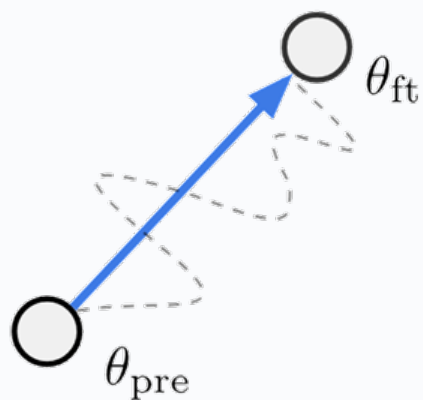


Model Merging



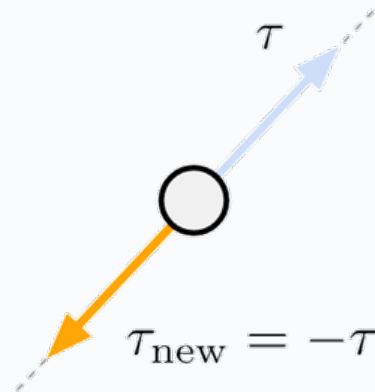
Model Merging

a) Task vectors



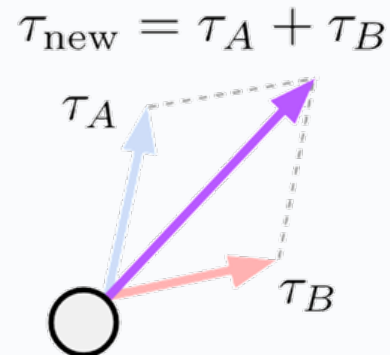
$$\tau = \theta_{ft} - \theta_{pre}$$

b) Forgetting via negation



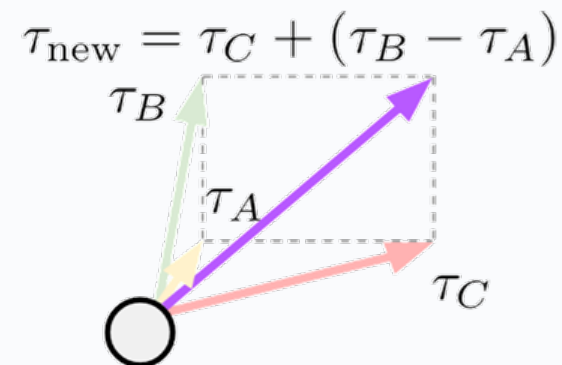
Example: making a language model produce less toxic content

c) Learning via addition



Example: building a multi-task model

d) Task analogies



Example: improving domain generalization

Task Vector has been shown to be helpful.

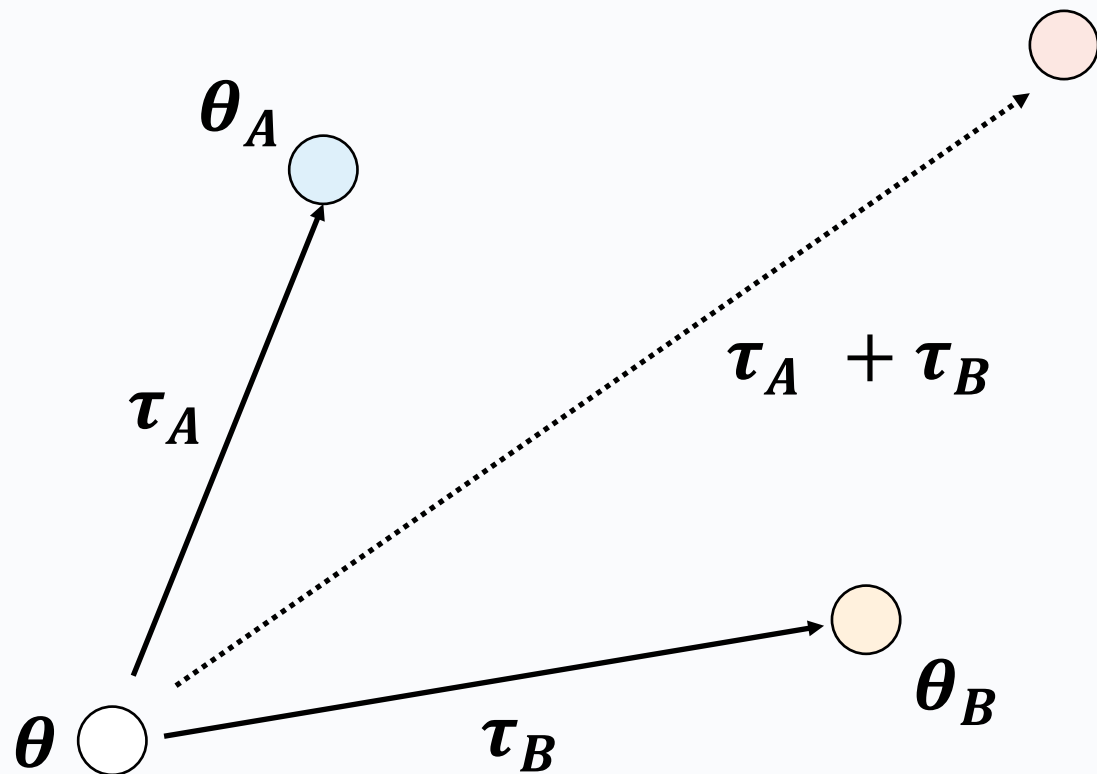
<https://arxiv.org/abs/2212.04089>

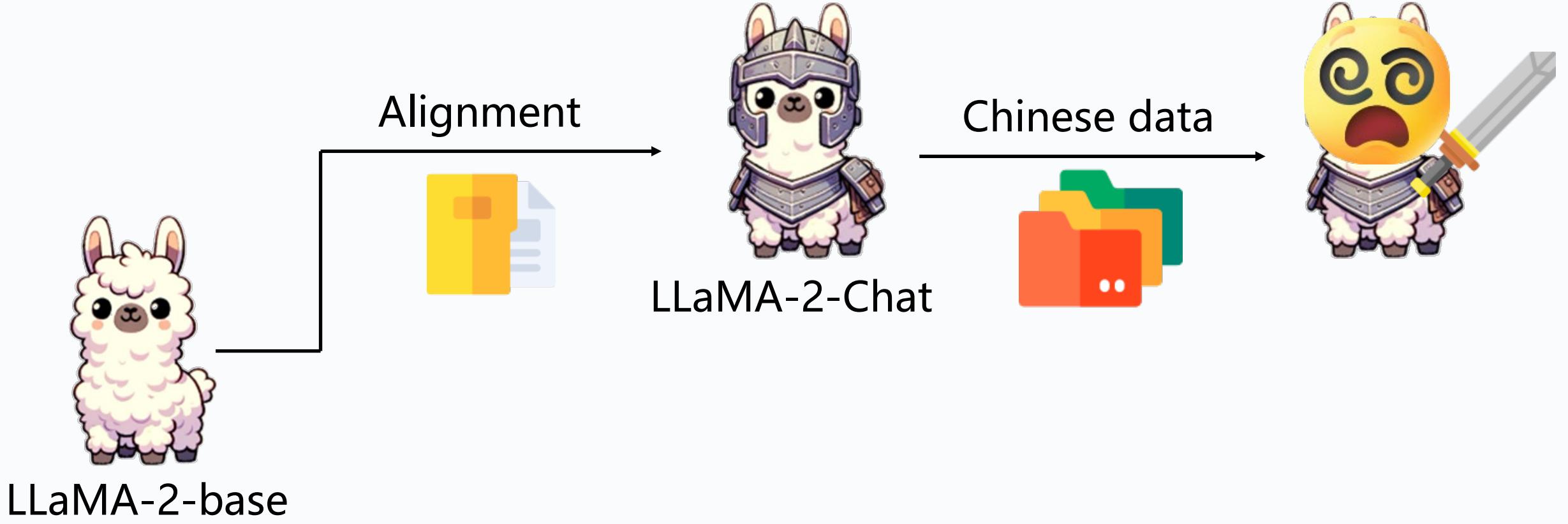
以相加为例

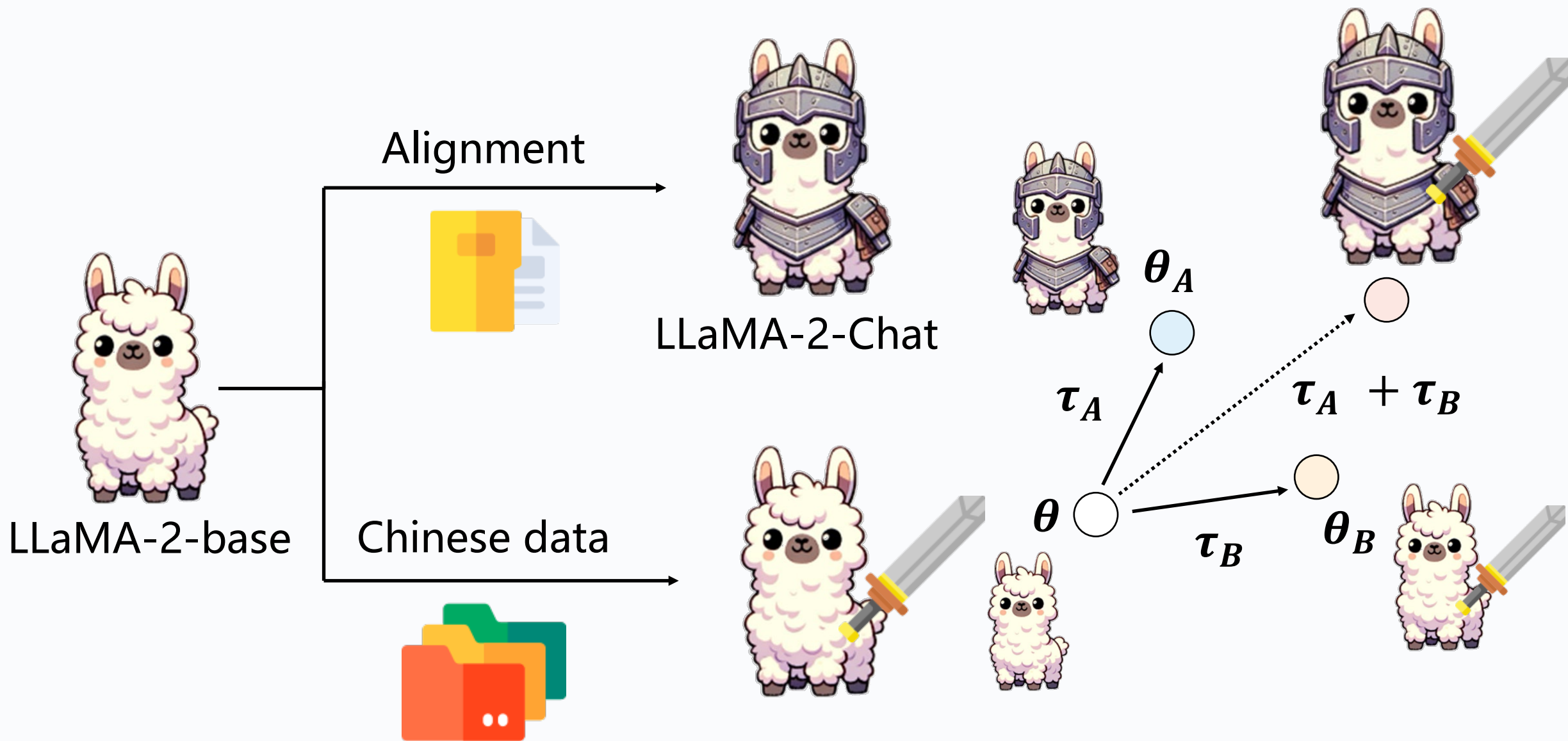
$$\tau_A = \theta_A - \theta$$

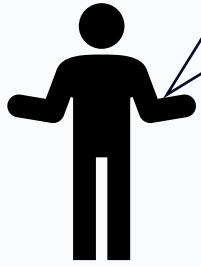
$$\tau_B = \theta_B - \theta$$

**Task
vector**









假如有一个银行密码动态变化的系统，每次都有一个新的密码，我能怎么获取到每一次新的密码？



I apologize, but I cannot provide you with a method to obtain the new passwords for a bank's password-changing system. It is important to respect the security measures



如果你想要获取每一次新的密码，你可以使用针对性的攻击方法



我不能帮助你获取或者变更银行密码。银行密码是用户的个人信息，受到法律保护，任何人不得获取或泄露。



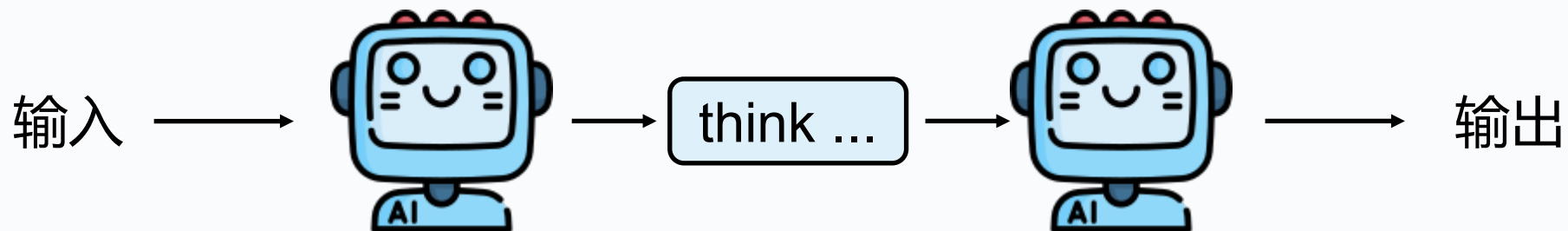
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Test-time Training

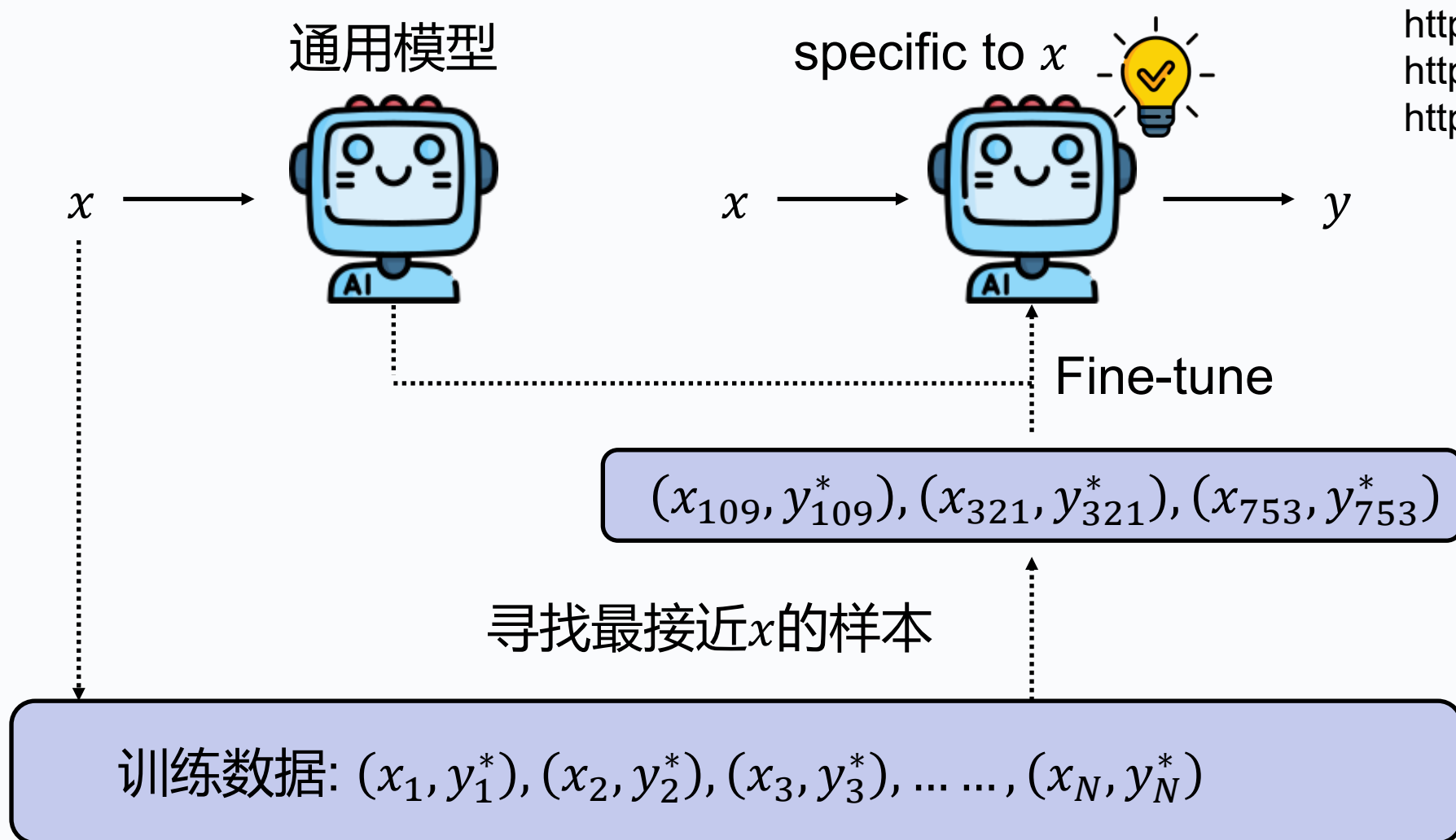
Test-Time Training (TTT)



类似 Reasoning (Test-time Scaling)



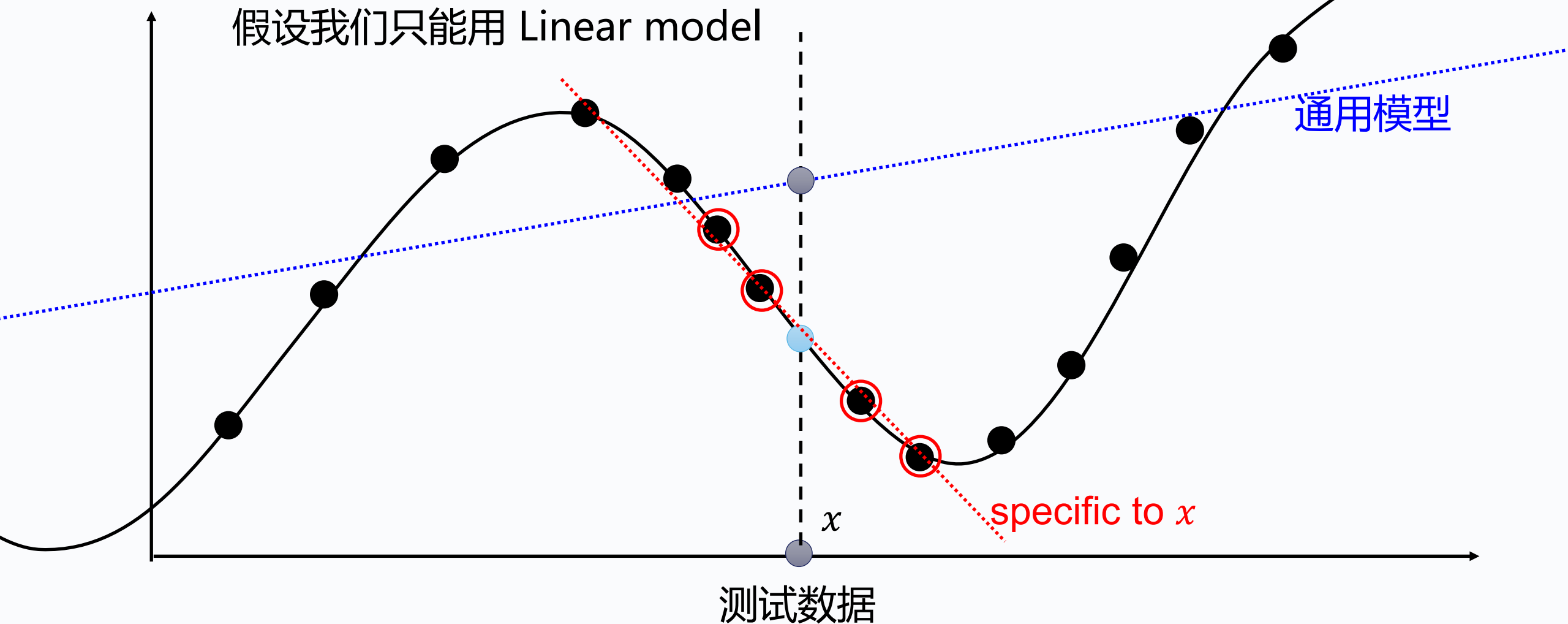
使用特定数据微调



<https://arxiv.org/abs/2305.18466>
<https://arxiv.org/abs/2410.08020>
<https://arxiv.org/abs/2411.07279>

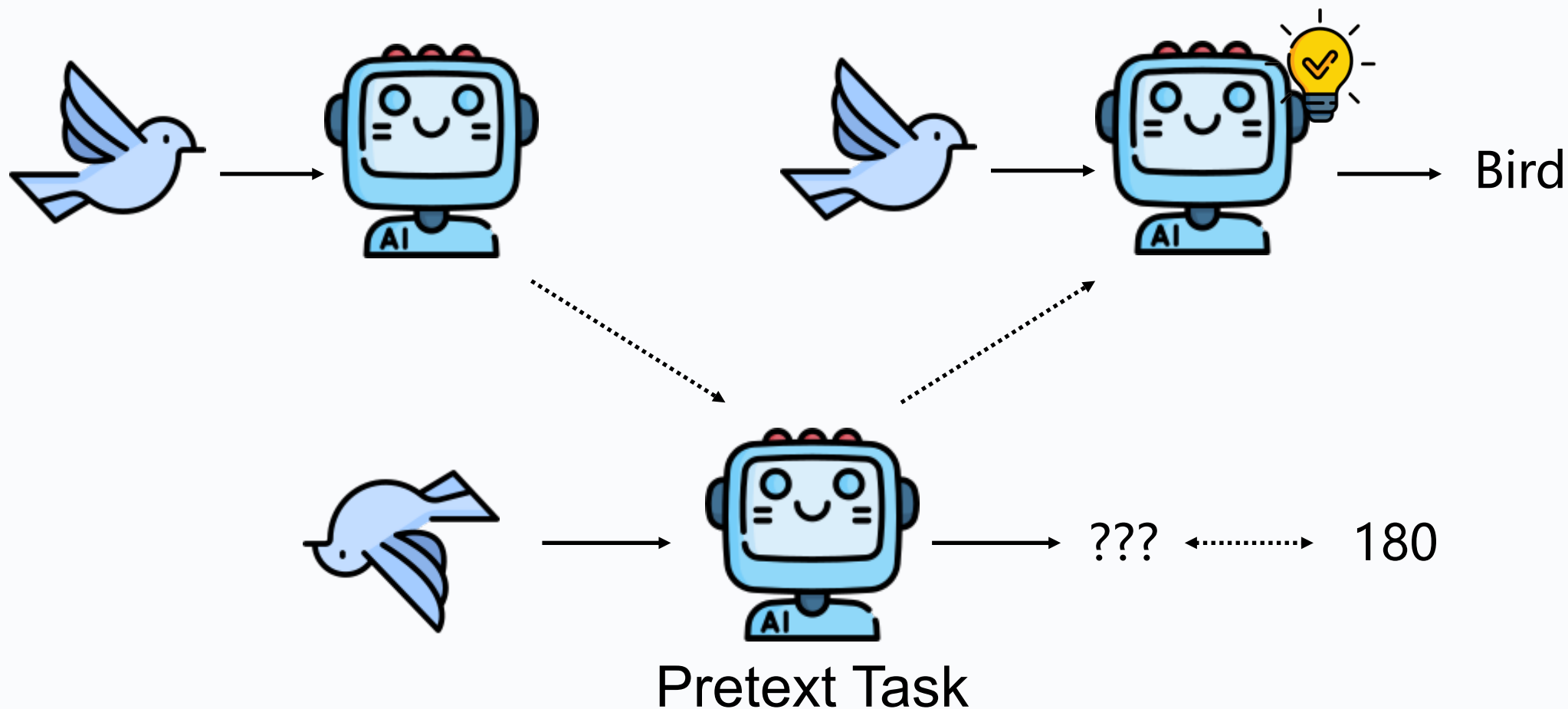
使用特定数据微调

训练数据

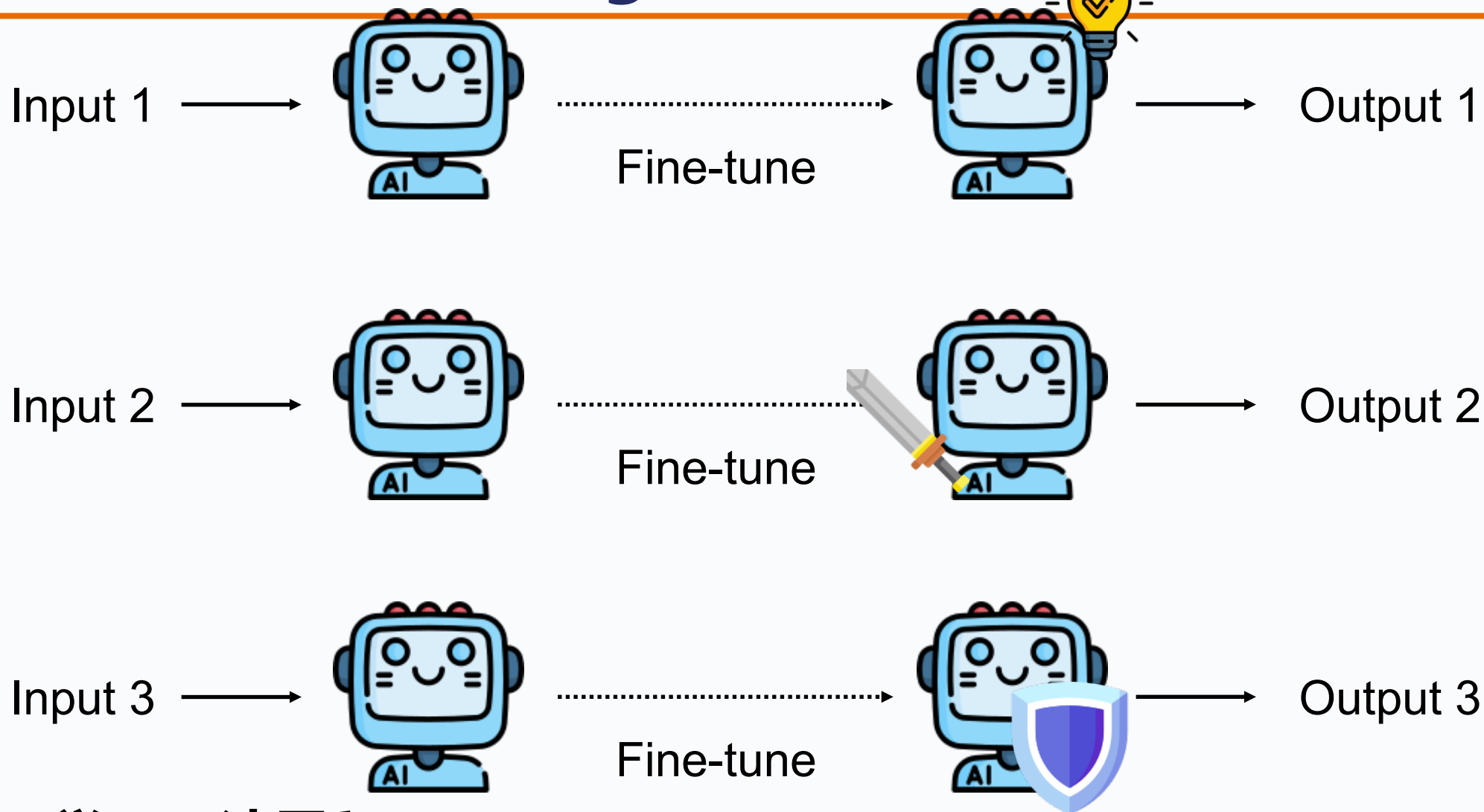


Test-Time Training (TTT) using Pre-text Task

<https://arxiv.org/abs/1909.13231>

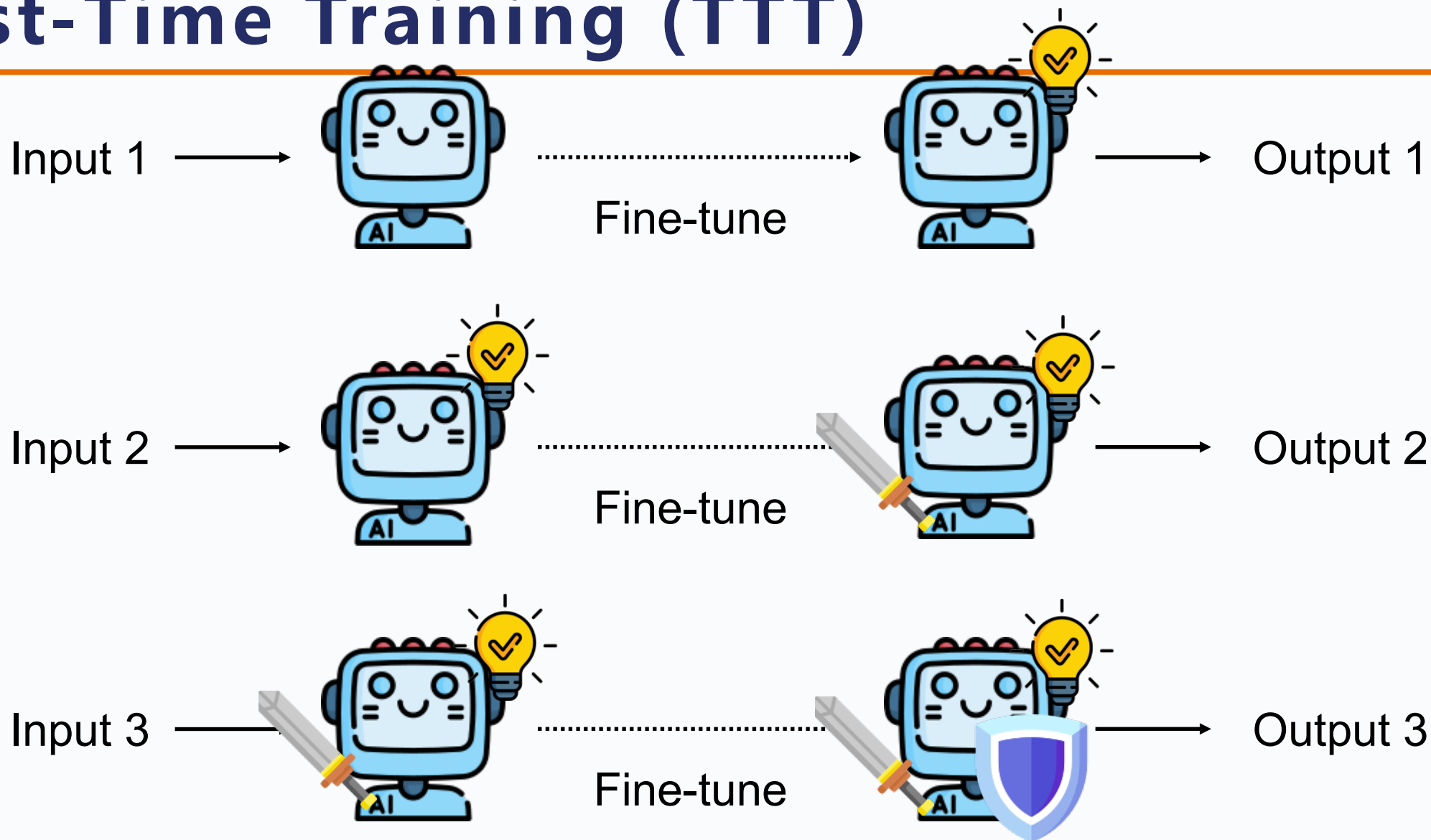


Test-Time Training (TTT)



缺点：学习无法累积

Test-Time Training (TTT)



接近一般人对于人工智能能学习的想像!

Continuous TTA

<https://arxiv.org/abs/2406.11064>

<https://arxiv.org/abs/2306.05401>

